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RESEARCH

The magnitude paradox

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Abstract
There is strong evidence indicating that relative risk aversion increases with outcome magnitude whereas impatience decreases with outcome magnitude. This finding seems paradoxical because it cannot be captured by the same utility function: Increasing relative risk aversion requires *decreasing elasticity* of the utility function, whereas decreasing impatience requires *increasing elasticity* with respect to outcome magnitudes. We develop a model that organically links the domains of risk taking and time discounting. The resulting two-speeds model generates a magnitude-dependent discount function such that increasing relative risk aversion is not only compatible with the magnitude effect in discounting but actually predicts magnitude-dependent discount weights. Moreover, we conduct a high-stakes laboratory experiment that reproduces the magnitude paradox and enables us to structurally estimate competing models of magnitude-dependent discounting. Both the Bayesian Information Criterion and the Akaike Information Criterion favor our two-speeds model. The results suggest that participants behave as if they apply a heuristic procedure: the value of a future reward consists of a certain percentage of the reward, irrespective of the length of delay, plus a delay-dependent component.

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1 Introduction

There is a large body of experimental evidence indicating that behavior under risk as well as behavior over time depend on outcome magnitude in systematic ways. In the domain of risk, decision makers' choices usually exhibit relative risk aversion *increasing* with outcome magnitude; in intertemporal choices, impatience *decreases* in outcome magnitude (Binswanger, 1981; Fehr-Duda et al., 2010; Frederick et al., 2002; Holt & Laury, 2002; Loewenstein & Thaler, 1989). These regularities pose a challenge to the existence of a general, domain-independent, utility function as the respective magnitude effects require opposing characteristics, decreasing elasticity with respect to risky monetary outcomes and increasing elasticity with respect to delayed outcomes. Because of this magnitude paradox many researchers have advocated domain-specific utility functions (Abdellaoui et al., 2013, 2011).

Such a solution to the magnitude paradox is less than satisfactory for several reasons. First of all, why should a dollar, conditional on its materialization, have a different instantaneous value in certain, risky, and delayed situations? Second, what kind of utility function applies when we are dealing with prospects that simultaneously involve risk and delay? Which one is driving behavior then, utility for risk or utility for time, or even a third variant of utility function? Finally, allowing for domain-specific utility generates additional degrees of freedom which the modeler would like to avoid.

We address this magnitude paradox by utilizing insights from economics, psychology, and neuroscience. First, an obvious way out of the dilemma is to locate the magnitude effect in discounting not in the utility function but in the discount weights, as proposed, for example, by Noor (2011) and Baucells and Heukamp (2012).

Second, given the prevalence of non-exponential discounting in experimental work - mostly manifest as impatience decreasing with time delay and labeled *hyperbolic discounting* (Kirby & Marakovic, 1995) - psychologists have studied different functional forms to describe discounting behavior (e.g. Mazur (1987)). Recently, researchers found that a mixture of exponential discount functions with differing discount rates provides a better fit than conventional specifications of hyperbolic discount functions (van den Bos & McLure, 2013). According to this finding, a future prospect $T = [x, t]$ promising a monetary reward x at time $t > 0$ is discounted by a mixture of two exponential discount weights: $D(t) = \omega \exp(-\eta t) + (1 - \omega) \exp(-\theta t)$, where $\eta \geq 0$ and $\theta \geq 0$ are two different discount rates and $0 < \omega < 1$ denotes the mixture weight. It is a well-known fact that mixtures of exponentials display hyperbolic discounting (e.g. Bell (1988)). These experimental findings are bolstered by contributions from neuroscience. One strand of the neuroscientific literature postulates, and presents evidence for, the existence of separate neural systems that are characterized by different degrees of impatience (Cavagnaro et al., 2016; McLure et al., 2007, 2004; van den Bos & McLure, 2013). Thus, we also adopt this evidence-based idea that discounting is driven by two exponential functions declining at different rates which is applicable whenever observed discounting behavior is hyperbolic.

The final building block of our model concerns the implementation of magnitude dependence of the discount weights. We modify the idea of mixing two exponentials by splitting the promised reward x into two components, δx and $(1 - \delta)x$ with

$0 < \delta < 1$, and applying a rank-dependent formula to the split: We assume that the utility $u(\delta x)$ is discounted at a different rate than the remaining utility $u(x) - u(\delta x)$. There are several possibilities of rationalizing such an assumption, embodied in the uncertainty of the future. For example, the decision maker may believe that part of the promised reward will materialize later than contracted for, or she may believe that it is likely that only a fraction of the promised reward will actually materialize at delay t. Thus, δ can be interpreted as a depreciation parameter and, together with the differing discount rates, may reflect devaluation for some kind of perceived future risk.

Overall, our model is based on three ideas, grounded in theoretical and empirical arguments:

1. The discount weights are the carriers of the magnitude effect. This assumption is derived from the observation that utility cannot display (strongly) decreasing elasticity and increasing elasticity at the same time.
2. Discounting is driven by a mixture of two exponentials declining at different rates. This principle is based on the empirical evidence from psychology and neuroscience, cited above.
3. A future reward is split into two components to which different discount rates are applied. This assumption constitutes the crucial new feature of our approach which we view as a reduced form of some underlying psychological process.

These principles generate a *magnitude-dependent two-speeds model* according to which a future prospect $T = [x, t]$ is evaluated as $u(\delta x)\exp(-\eta t) + [u(x) - u(\delta x)]\exp(-\theta t)$. Thus, the model features three potential sources of devaluing a future reward: the depreciation parameter $0 < \delta < 1$ and the two differing discount rates $\eta \geq 0$ and $\theta \geq 0$. Our key insight is that, in this framework, decreasing elasticity of the utility function implies a magnitude effect of the desired direction for the discount weights. In other words, relative risk aversion increasing in outcome magnitude predicts the existence of a magnitude effect in time discounting. Thus, our two-speeds model is consistent with risk taking and time discounting behaviors being driven by the same domain-independent utility function. Furthermore, our approach engenders hyperbolic discounting without resorting to any additional assumption, as often required, and, given the specifics of the utility function, prescribes clearly and unambiguously the functional specification of the discounting model.

We test our predictions by estimating a structural model on the basis of novel experimental data collected on the premises of a European university. We elicited both certainty equivalents for a large set of risky prospects and sooner equivalents for delayed prospects involving comparatively high real stakes. Our data exhibits the following characteristics:

- Relative risk aversion increases with outcome magnitude,
- impatience decreases with time delay, i.e., hyperbolic discounting, and
- impatience decreases with outcome magnitude, i.e., the classical magnitude effect.

Thus, our data displays the very combination of features that our two-speeds model is meant to accommodate. Structural estimates show that indeed, in the risk domain, a utility function exhibiting decreasing elasticity fits the data best which we subsequently use for estimating the parameters δ, η, and θ of the discount functional. Results at the individual level corroborate our key prediction that decreasing elasticity of utility implies the magnitude effect in discounting: We find a highly significant relationship between the measures for decreasing elasticity of the *estimated* individual utility functions and the magnitude effects in the *observed* individual discount rates. Moreover, the aggregate two-speeds model performs best in capturing observed discounting behavior when compared with other candidate models of magnitude-dependent discounting suggested in the literature. Interestingly, only two of the three potential sources of devaluing a future reward are active in capturing discounting behavior: The estimated discount rate η attached to $u(\delta x)$ turns out to be zero, i.e., $u(\delta x)$ is not discounted at all, whereas the value of the remaining reward $u(x) - u(\delta x)$ is discounted according to the length of delay. It seems as if the participants devalue a future reward by some heuristic procedure: first devaluing the future reward by δ for being delayed *per se* and then discounting the remaining utility according to the length of delay. Our results also lend themselves to an alternative interpretation that links participants' behavior to their perception of future uncertainty. Participants behave as if they expect to receive the promised reward at t with a certain delay-dependent probability, and only a fraction δ of the reward otherwise. The result of $\eta = 0$ is interesting in the light of previous neuroscientific evidence by McLure et al. (2007) who apply a mixture of exponentials to their data and also find that the smaller of the two discount rates is indistinguishable from zero.

The remainder of the paper is structured as follows. In Section 2, we explain our modeling approach in detail. Sections 3 and 4 are dedicated to the experimental design and data. Our econometric approach is set out in Section 5, followed by the estimation results in Section 6. Results for behavior at the individual level are summarized in Section 7. Finally, Section 8 concludes. Complementary materials are available in the Appendix.

2 The two-speeds model

Before we present the specifics of the discount functional, we explain the magnitude paradox, i.e., why the utility function in a standard model would need to exhibit opposing characteristics to be compatible with the magnitude effects in risk taking and time discounting.

2.1 The magnitude paradox

Consider first two risky prospects (x, p) and (y, q), paying monetary amounts x with probability p and y with probability q, respectively, where $x > y > 0$, $p < q$ and $(x, p) \sim (y, q)$. The magnitude effect in risk taking emerges for $\lambda > 1$ if

$$(\lambda x, p) \prec (\lambda y, q),$$

implying

$$\frac{u(\lambda x)}{u(\lambda y)} < \frac{u(x)}{u(y)},$$

which is equivalent to u exhibiting decreasing elasticity in a standard separable representation, i.e., where the value of a prospect (x, p) is given by $u(x)p$ (or $u(x)w(p)$, potentially including probability distortions $w(p)$) (Bouchouicha & Vieider, 2017).

In the domain of time discounting, the argument runs as follows. Consider two temporal prospects $[x, t_2]$ and $[y, t_1]$, paying monetary amounts x at delay t_2 and y at delay t_1, with $[x, t_2] \sim [y, t_1]$ and $x > y > 0$, $t_2 > t_1$. The magnitude effect in discounting appears for $\lambda > 1$ if

$$[\lambda x, t_2] \succ [\lambda y, t_1],$$

implying

$$\frac{u(\lambda x)}{u(\lambda y)} > \frac{u(x)}{u(y)},$$

which means that u is characterized by increasing elasticity, as has already been noted by Loewenstein and Prelec (1992).

2.2 The discount functional

Consider a temporal prospect $T = [x, t]$, $x > 0$, $t > 0$. The function u denotes the utility of monetary amounts x, assumed to be monotonically increasing and twice differentiable. As already described in the Introduction, we posit that the promised reward x is split into two components, δx and $(1-\delta)x$, $0 < \delta < 1$, that are discounted at different rates $\eta \geq 0$ and $\theta \geq 0$, respectively. Assume, without loss of generality, that $\theta > \eta$. Thus, the component $(1-\delta)x$ is discounted at a higher rate than δx, which can be interpreted as if $(1-\delta)x$ materializes with an additional delay $\Delta t = \frac{\theta-\eta}{\eta}t$ (for $\eta \neq 0$). Blavatskyy (2016) presents a theoretical justification that, in such a situation, incremental utility is the correct modeling choice.[1] In other words, the resulting temporal prospect $[(\delta x, t), ((1-\delta)x, t+\Delta t)]$ should be evaluated as

$$\begin{aligned} V_0[(\delta x, t), ((1-\delta)x, t+\Delta t)] &= u(\delta x)\exp(-\eta t) + [u((1-\delta)x + \delta x) - u(\delta x)]\exp(-\eta(t+\Delta t)) \\ &= u(\delta x)\exp(-\eta t) + [u(x) - u(\delta x)]\exp(-\theta t). \end{aligned}$$

[1] Using incremental utility $u(x) - u(\delta x)$ rather than $u((1-\delta)x)$ is the preferred model here as the latter implies under concave utility that splitting the reward x into δx and $(1-\delta)x$ may lead to preferring the split rather than the total of x (due to Jensen's inequality). Blavatskyy and Maafi (2018) find that rank-dependent discounting provides the best fit for the majority of the participants in their experiment.

Consequently, we assume such a rank-dependent formula for evaluating $T = [x, t]$. The present value of the prospect $V_0(T)$ is then given by

$$V_0(T) = u(\delta x)\exp(-\eta t) + [u(x) - u(\delta x)]\exp(-\theta t). \tag{1}$$

Define $\omega(x) := \frac{u(\delta x)}{u(x)}$. Rearranging terms in (1) yields

$$\begin{aligned} V_0(T) &= u(x)[\omega(x)\exp(-\eta t) + (1-\omega(x))\exp(-\theta t)] \\ &= u(x)D(t;x), \end{aligned} \tag{2}$$

where, for $\theta > \eta$, $0 < \omega(x) < 1$ is the magnitude-dependent mixture weight and $D(t;\, x)$ denotes the discount functional.[2] Consequently, the discount functional is a convex combination of two exponentials that decline at different rates where the mixture weight $\omega(x)$ depends not only on the magnitude of the outcome x, but also on the depreciation parameter δ and the characteristics of the utility function. If $\delta = 0$ ($\delta = 1$, $\eta = \theta$), the model collapses to the standard exponential discounting model. Another special case is generated by linear utility. In this case, $\omega(x) = \delta$ is constant. Moreover, since $\omega(x) > \delta$ for strictly concave utility functions, an individual with linear utility will appear less patient ceteris paribus, and will not display a magnitude effect in discounting. Furthermore, Bell (1988) shows that the resulting discount functional $D(t) = \omega\exp(-\eta t) + (1-\omega)\exp(-\theta t)$ with $\eta, \theta > 0$, exhibits the *one-switch property*.[3]

As many authors have demonstrated, a mixture of exponential discount functions generates decreasing impatience, i.e., hyperbolic discounting (Bell, 1988; Gollier & Zeckhauser, 2005; Jackson & Yariv, 2015; Weitzman, 2001). Prelec (2004) extends this insight to discount functions that may be decreasingly impatient right from the outset and demonstrates that a mixture of two different decreasingly impatient discount functions represents more decreasingly impatient preferences. He further notes that the function with the higher rate of time preference controls the shape of the mixed function in the early time periods, while the function with the lower rate of time preference takes over for the more distant periods. His result can also be used to compare mixtures: a 50–50 mixture will be more decreasingly impatient than a 25–75 mixture, for example. Prelec's insights indicate that, in the two-speeds model, outcome magnitude and the degree of hyperbolic decline of the overall discount rate are intimately intertwined.

The most interesting implication of our approach concerns the behavior of $D(t;\, x)$ with respect to changes in outcome magnitude. It turns out that the characteristics of

[2] The same logic applies to the case of $\theta < \eta$. Define $\tilde{\delta} := 1 - \delta$ and $\tilde{\omega}(x) := \frac{u(\tilde{\delta}x)}{u(x)}$. Then it is straightforward to show that the discount functional is given by $\tilde{D}(t;x) = \tilde{\omega}(x)\exp(-\theta t) + (1-\tilde{\omega}(x))\exp(-\eta t)$. Thus, the mixture weight $\tilde{\omega}(x)$ is again combined with the *lower* discount rate (and $1 - \tilde{\omega}(x)$ with the higher discount rate), which is a consequence of the rank-dependent formulation and does not constitute an additional assumption.

[3] The one-switch property holds if adding a common delay to two sequences of dated outcomes leads to preference reversals exactly once: if $[x, t] \succ [y, s]$, then there exists a unique delay d^*, such that $[y, s + d] \succ [x, t + d]$ for all $d > d^*$.

the elasticity $\varepsilon(x)$ of $u(x)$ are crucial drivers for the occurrence of a magnitude effect in discounting.

PROPOSITION: Magnitude Dependence of D (t ; x)

Given $x > 0$ and $\theta > \eta \geq 0$, then

$$\frac{\partial D(t;x)}{\partial x} \begin{cases} > 0 & \text{if } \varepsilon(x) \text{ decreasing} \\ = 0 & \text{if } \varepsilon(x) \text{ constant} \\ < 0 & \text{if } \varepsilon(x) \text{ increasing} \end{cases}$$

Proof of Proposition

Rearranging the terms of $D(t; x)$ yields

$$\begin{aligned} D(t;x) &= \omega(x)\exp(-\eta t) + (1-\omega(x))\exp(-\theta t) \\ &= \omega(x)(\exp(-\eta t) - \exp(-\theta t)) + \exp(-\theta t). \end{aligned}$$

Note that $\exp(-\eta t) - \exp(-\theta t) > 0$ as $\theta - \eta > 0$. Therefore, the derivative of $D(t; x)$ with respect to x depends on the sign of the derivative of $\omega(x) = \frac{u(\delta x)}{u(x)}$.

$$\begin{aligned} \frac{\partial}{\partial x}\left[\frac{u(\delta x)}{u(x)}\right] &= \frac{u(x)u'(\delta x)\delta - u'(x)u(\delta x)}{[u(x)]^2} \\ &= \frac{u(\delta x)}{xu(x)}\left[\frac{u'(\delta x)\delta x}{u(\delta x)} - \frac{u'(x)x}{u(x)}\right] \\ &= \frac{u(\delta x)}{xu(x)}\left[\varepsilon(\delta x) - \varepsilon(x)\right]. \end{aligned}$$

If the elasticity of the utility function decreases in x, then $\varepsilon(\delta x) - \varepsilon(x) > 0$ and, therefore, $\frac{\partial D(t;x)}{\partial x} > 0$ ■

Thus, the existence of a magnitude effect in the discount weights depends on the elasticity of $u(x)$. If the elasticity is decreasing, capturing increasing relative risk aversion, then the magnitude effect in the discount weights appears.

Figure 1 illustrates the magnitude effect in the overall utility discount rate $\rho(t;x) := -\frac{\partial D(t;x)/\partial t}{D(t;x)}$ when $\theta > \eta$ and the utility function is characterized by decreasing elasticity. For illustrative purposes we use a suitable functional form, the hybrid expo-power specification, which we will also use later for the empirical analysis. This specification, a variant of Abdellaoui et al. (2007)'s suggested functional form, displays decreasing elasticity under appropriate parameter restrictions:

$$u(x) = 1 - \exp(-\frac{x^\gamma}{\gamma}). \tag{3}$$

In particular, for $0 < \gamma < 1$, the expo-power function is concave and has the desirable features accommodating increasing relative risk aversion and decreasing absolute risk aversion at the same time. Therefore, in our model, a utility function with decreasing elasticity accommodates both relative risk aversion increasing with outcome magnitude and discount weights increasing with outcome magnitude. The

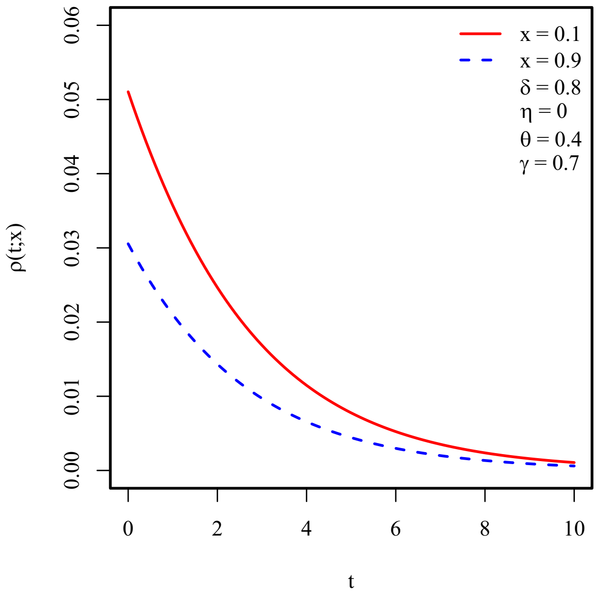


Fig. 1 Magnitude Effect in Discounting Behavior. The graph shows overall utility discount rates $\rho(t;x) = -\frac{\partial D(t;x)/\partial t}{D(t;x)}$ as they move with the length of delay t for different levels of the outcome $x \in \{0.1, 0.9\}$, where $u(x) = -\exp(-x^{\gamma}/\gamma) + 1$, with $\gamma = 0.70$. The depreciation parameter δ is assumed to be 0.8, the component discount rates amount to $\theta = 0.4$ and $\eta = 0$, respectively

overall discount rates $\rho(t;x)$ in Fig. 1 show both, a magnitude effect and a hyperbolic decline, as predicted.

2.3 Extension of the two-speeds model

While a mixture of exponentials may well be superior to other models of discounting at the level of aggregate behavior, it must fail for participants who exhibit increasing impatience. In fact, many experimenters have found a non-negligible percentage of their participants to exhibit such a pattern. In order to accommodate increasing impatience as well, we suggest to adapt our model of $D(t;x)$ in the following way:

$$D(t;x) = \omega(x)\exp(-(\eta t)^{\kappa}) + (1-\omega(x))\exp(-(\theta t)^{\kappa}), \tag{4}$$

based on constant-sensitivity discount functions with an additional parameter κ. Such a functional form, introduced by Ebert and Prelec (2007), can accommodate increasing impatience whenever $\kappa > 1$. We will also put this extended version of our model to the test.

When the utility function exhibits decreasing elasticity and $\theta \geq \eta$, our extended model can map the following patterns:

1. Decreasing impatience plus magnitude effect ($\theta > \eta$ and $\kappa \leq 1$)
2. Increasing impatience plus magnitude effect ($\theta > \eta$ and $\kappa > 1$)
3. Decreasing impatience without magnitude effect ($\theta = \eta$ and $\kappa \leq 1$)
4. Increasing impatience without magnitude effect ($\theta = \eta$ and $\kappa > 1$)
5. Constant discounting without magnitude effect ($\theta = \eta$ and $\kappa = 1$)

Thus, our model fails at capturing magnitude-dependent *constant* discounting.

2.4 Relation to previous literature

Two prominent contributions deal with magnitude effects in discounting, Noor (2011) and Baucells and Heukamp (2012). Noor (2011) provides axiomatic foundations for a magnitude-dependent discounting model by weakening the classical stationarity axiom of Discounted Utility Theory. He derives a representation

$$V_0([x,t]) = u(x)D(t;x) = u(x)d(x)^t, \tag{5}$$

where the discount factor $d(x)^t$ depends on outcome magnitude with $d'(x) > 0$. Thus, the discount factor is *exponential* for a given magnitude of x, but increases with x. The resulting generalized class of exponential discount functions encompasses countless possible functional relationships between discount weights and outcome magnitude. The discount factor may well depend on $u(x)$ but Noor (2011) clarifies that risk attitude remains independent of time.

If the data exhibits such a pattern,[4] a suitable parameterization of Noor (2011) may be the model of choice, or serving as a component in a finite mixture model (McLachlan & Peel, 2000) when there is heterogeneity with respect to discounting types.[5] Since the two-speeds model cannot capture magnitude effects when discounting is exponential, Noor (2011) constitutes a valuable complement to our approach. For empirical work, specifications of the utility function and of the type of magnitude dependence of the discount factor are required. Using Noor (2011)'s model for our data actually constitutes a case of mis-specification as the data displays a pronounced pattern of hyperbolic discounting. Nevertheless, for the sake of completeness, we present estimation results in Appendix C.

In their axiomatic model Baucells and Heukamp (2012) consider preferences over triplets (x, p, t) that describe the prospect of receiving x with probability p at delay t, zero otherwise. The crucial axiom is their so-called *subendurance* which assumes that the larger the reward, the more willing subjects are to wait in exchange for improved probabilities. Thus, magnitude dependence is directly built into the model assumptions. In our setting, where $p = 1$, their representation boils down to

$$V_0([x,t]) = u(x)\exp(-d(\rho(x)t)), \tag{6}$$

where $\rho(x)$, $\rho'(x) < 0$, denotes the magnitude-dependent discount rate and $d(\tau)$ is a continuous strictly increasing function with $d(0) = 0$ and $lim_{\tau\to\infty} d(\tau) = \infty$. $d(\tau)$ is labeled *psychological distance function*. Thus, the model can accommodate both exponential and non-exponential discounting behaviors depending on the char-

[4] A simple way to uncover magnitude-dependent *exponential* discounting in experimental data is based on the elicitation of *delay equivalents*, instead of sooner equivalents as in our data set. A description of such a procedure is included in Appendix C.

[5] Applications of finite mixture models for identifying different risk taking types can be found in Bruhin et al. (2010) and Conte et al. (2010).

acteristics of $d(\tau)$. For $d(\tau) = \tau$, $V_0([x, t]) = u(x) \exp(-\rho(x)t)$ results which is equivalent to the magnitude-dependent model by Noor (2011). If $d(\tau) = \tau^\kappa$ with $0 < \kappa < 1$, d is concave, and we end up with $V_0([x, t]]) = u(x) \exp(-(\rho(x)t)^\kappa)$, which corresponds to Ebert and Prelec (2007)'s functional form mapping hyperbolic discounting. When working with such a framework, the experimenter will therefore have to specify functional forms of several model components: the utility function $u(x)$, the type of magnitude dependence of the discount rate $\rho(x)$, and the psychological distance function $d(\tau)$. In other words, there is *a priori* no direct link between utility and discount weights.

Baucells and Heukamp (2012) are well aware of the magnitude paradox and explictly note that they *"attribute the magnitude effect to subendurance, which, if sufficiently strong, can offset the possible effect of a decreasing elasticity of [u]. In fact, if the elasticity of [u] is constant, as often assumed, then"* we observe a magnitude effect in discounting iff subendurance holds (page 836). Of course, constant elasticity of utility precludes a magnitude effect in risk taking and, therefore, provides no solution to the magnitude paradox.

Since their model can accommodate hyperbolic discounting it is a suitable contender for the two-speeds model. Obviously it imposes a completely different structure on the data. Ultimately, it is an empirical question which one of the two models captures the behavioral patterns in the data better. We will discuss model estimates for Baucells and Heukamp (2012) in Section 6.

3 The experiment

In order to test our model, we conducted a laboratory experiment with a number of distinguishing features:[6]

1. We conducted the experiment with 192 students on the premises of the University of Piraeus in Athens, Greece, in 2016.
2. The experiment involved both risk taking and time discounting tasks. Each participant faced all the tasks.
3. The risk taking data are sufficiently rich to separately identify utility curvature and probability weighting.[7]
4. Stake sizes in both decision domains were sufficiently large to detect magnitude effects.
5. The experiment was fully incentivized, i.e., each single participant was paid in an incentive compatible manner.[8]
6. The payment procedures were set up in a way to signal maximum safety in order to prevent mistrust in the experimenters' reliability.
7. We kept transaction costs constant across the different payment dates.

[6] Experimental instructions are available in the Online Appendix.

[7] This is important because the risk taking data exhibit substantial probability dependence.

[8] In many experiments on time discounting and/or substantial stakes only a few participants are paid, which generates an additional layer of riskiness on the experimental tasks. We aimed at avoiding such a confound.

8. Finally, the relatively high stakes allowed us to denominate the payoffs in EUR rather than in experimental currency units.

The Department of Banking and Financial Management hosted the experiment and made the dean's office available for the payment procedures. The experiment was run on computers with the software z-Tree (Fischbacher, 2007). Conducting the experiment in Athens offered the advantage that we could pay much higher amounts than we could have paid at our home universities. On average, participants earned about seven times as much in real terms compared to typical participants' earnings at our home universities.

As mentioned, the experiment consisted of two tasks, a risk taking task and a time discounting task. We randomized the order of the two tasks to avoid order effects. The written instructions, available in the Online Appendix, were in English, and fluency in English was a prerequisite for participation. In addition, before the experiment started, a verbal introduction was given in Greek always by the same person.

Participants could work at their own speed during the entire experiment and needed on average 75 minutes to complete it. Choices were fully incentivized, i.e., each participant received a payment for one randomly drawn decision in the risk taking task and one randomly drawn decision in the time discounting task. Aside from a show-up fee of EUR 10, average earnings amounted to EUR 56.40 for the risk taking task, paid out immediately after completion of the respective sessions, and EUR 55.90 for the time discounting task. To keep transaction costs for the discounting task constant across payment dates, the earliest possible payment date was the day after the experiment when participants could pick up their payments at the dean's office. All the remaining delayed payments were made at that location by a doctoral student who also helped with the payments resulting from the risk taking task paid out after the sessions. Thus, he was familiar to the participants. Aside from a written confirmation of the amount and timing of the experimental earnings, the doctoral student remained the participants' contact person and reminded them by email when their payments were about to be due. The payment details were explained carefully at the beginning of each session to disperse any doubts about payment safety.

3.1 The risk taking task

In the risk taking task, we elicited certainty equivalents for 36 binary risky prospects, $R = (x, p; y, 1 - p)$, with $x > y \geq 0$. As we wanted to create a finely grained map of participants' risk preferences, the design of the risky prospects allowed us to check for changes in relative as well as in absolute risk aversion. The latter dimension has received comparatively little attention in the experimental literature but is important for discriminating among different specifications of the utility function.

Table 1 summarizes the design of the 36 risky prospects. We first defined 9 *base prospects* with an expected value of EUR 13.50 that vary in the probability of the best outcome (10%, 50% and 90%) and standard deviation (1.5, 3 and 4.5). Subsequently, we constructed 9 *medium shift prospects* and 9 *large shift prospects* by adding EUR 20 and EUR 40, respectively, to the outcomes of each base prospect. A comparison of the certainty equivalents of the base prospects with the certainty equivalents of

Table 1 Design of the 36 Risky Prospects

	Base			Medium Shift			Large Shift			Scaled		
ID	x	p	y	x	p	y	x	p	y	x	p	y
1	18.0	0.1	13.0	38.0	0.1	33.0	58.0	0.1	53.0	90.0	0.1	65.0
2	22.5	0.1	12.5	42.5	0.1	32.5	62.5	0.1	52.5	112.5	0.1	62.5
3	27.0	0.1	12.0	47.0	0.1	32.0	67.0	0.1	52.0	135.0	0.1	60.0
4	15.0	0.5	12.0	35.0	0.5	32.0	55.0	0.5	52.0	75.0	0.5	60.0
5	16.5	0.5	10.5	36.5	0.5	30.5	56.5	0.5	50.5	82.5	0.5	52.5
6	18.0	0.5	9.0	38.0	0.5	29.0	58.0	0.5	49.0	90.0	0.5	45.0
7	14.0	0.9	9.0	34.0	0.9	29.0	54.0	0.9	49.0	70.0	0.9	45.0
8	14.5	0.9	4.5	34.5	0.9	24.5	54.5	0.9	44.5	72.5	0.9	22.5
9	15.0	0.9	0.0	35.0	0.9	20.0	55.0	0.9	40.0	75.0	0.9	0.0

the shift prospects directly reveals stake effects on absolute risk aversion (Farquhar & Nakamura, 1987). Finally, we created 9 *scaled prospects* by multiplying the outcomes of the base prospects by 5. The certainty equivalents for these prospects give immediate insight into magnitude effects in relative risk aversion, the ultimate target of our investigation.

We elicited the certainty equivalents with ordered choice menus. Figure 2 shows an example of such a choice menu for a risky prospect. In each of the 21 rows of the choice menu, participants had to indicate whether they preferred the fixed payment (Option A) or the risky prospect (Option B). The fixed payment varied in equally sized steps from the lowest to the highest outcome of the risky prospect. We define the observed certainty equivalent, *ce*, as the mean of the two fixed payments where the participant switched from preferring the fixed payment to preferring the risky prospect. We did not enforce monotonicity, but Option A was always pre-selected by default. However, we dropped observations of participants' choices when they violated first order stochastic dominance (by choosing the fixed payment in the bottom row) or entered multiple switching points. All of a participant's choices were dropped when she committed these types of errors in more than three choice menus overall.

3.2 The time discounting task

In the time discounting task, we elicited 24 sooner equivalents $se(x)$ payable at t_1 for temporal prospects, $T = [x, t_2]$, $t_2 > t_1$. Table 2 summarizes the design of the temporal prospects. There were six combinations between the delay t_1, ranging from 0 to 4 months, of the sooner equivalent $se(x)$, and the later delay of the future payment x, t_2, which varied from 2 to 6 months. For each of these six combinations, the payment of the temporal prospect, T, amounted to either EUR 14, 34, 67, or 135.

The time discounting task is based on a classical smaller-sooner versus larger-later design. Similar to the risk taking task, we elicited sooner equivalents $se(x)$ of given later payments x with ordered choice menus. Figure 3 shows an example of such a choice menu. In each of the 21 rows, participants had to choose between a sooner payment with a delay t_1 (Option A) and the payment of x with a delay t_2 (Option B). The sooner payment varied in equally sized steps from EUR 0 to x. The observed sooner equivalent, $se(x)$, is defined as the mean of the two sooner payments where the participant switched from preferring the sooner payment to preferring the later payment. Option A was always pre-selected by default. As in the risk taking task, we did not enforce monotonicity. However, we dropped observations of participants who either chose the sooner payment of zero in the bottom row or entered multiple switching points. Participants' observations were completely dropped when they committed these types of errors in more than three choice menus overall. Applying this procedure across all choice menus for risk and time resulted in a total of 33 dropouts (17%), leaving us with 159 participants (59% male).[9] Average age amounted to 21 years.

[9] Our analysis is robust with respect to the number of violations used as exclusion criterion.

Option A Fixed Payment	Your Choice		Option B Gamble
54.50	● A	○ B	54.50 with a chance of 90% 44.50 otherwise (a chance of 10%)
54.00	● A	○ B	
53.50	● A	○ B	
53.00	● A	○ B	
52.50	● A	○ B	
52.00	● A	○ B	
51.50	● A	○ B	
51.00	● A	○ B	
50.50	● A	○ B	
50.00	○ A	● B	
49.50	○ A	● B	
49.00	○ A	● B	
48.50	○ A	● B	
48.00	○ A	● B	
47.50	○ A	● B	
47.00	○ A	● B	
46.50	○ A	● B	
46.00	○ A	● B	
45.50	○ A	● B	
45.00	○ A	● B	
44.50	○ A	● B	

Fig. 2 Choice Menu for Eliciting the Certainty Equivalent of a Risky Prospect. In each row participants had to indicate whether they preferred the respective certain payment ("Option A, Fixed Payment") or the risky prospect ("Option B, Gamble"). The observed certainty equivalent *ce* is calculated as the average of the two fixed payments where a preference switch occurred. In this example, the observed certainty equivalent is EUR 50.25

Table 2 Design of the 24 Temporal Prospects

Delays in months		Payment x of the temporal prospect in EUR			
Sooner t_1	Later t_2				
0	2	14	34	67	135
0	4	14	34	67	135
0	6	14	34	67	135
2	4	14	34	67	135
2	6	14	34	67	135
4	6	14	34	67	135

4 Descriptive analysis

This section gives a descriptive overview of the experimental data. It uses model-free measures to summarize the main patterns in participants' risk taking and time discounting behaviors, measured by the observed certainty equivalents and sooner equivalents, respectively.

4.1 Risk taking data

To measure participants' attitudes towards the risky prospect R, we rely on the risk premium $RP(R) = ev(R) - ce(R)$, where $ev(R)$ denotes the risky prospect's expected outcome and $ce(R)$ its certainty equivalent. If the risk premium is zero, the decision maker is risk neutral. A positive risk premium indicates risk aversion, a negative one indicates risk seeking. Normalizing $RP(R)$ by the absolute value of the expected outcome yields the relative risk premium,

	Option A in 0 months + 1 day	Your Choice		Option B in 4 months + 1 day
1	66.00	◉ A	○ B	
2	62.70	◉ A	○ B	
3	59.40	◉ A	○ B	
4	56.10	◉ A	○ B	
5	52.80	◉ A	○ B	
6	49.50	◉ A	○ B	
7	45.20	◉ A	○ B	
8	42.90	◉ A	○ B	
9	39.60	◉ A	○ B	
10	36.30	◉ A	○ B	
11	33.00	◉ A	○ B	66.00
12	29.70	○ A	◉ B	
13	26.40	○ A	◉ B	
14	23.10	○ A	◉ B	
15	19.80	○ A	◉ B	
16	16.50	○ A	◉ B	
17	13.20	○ A	◉ B	
18	9.90	○ A	◉ B	
19	6.60	○ A	◉ B	
20	3.30	○ A	◉ B	
21	0.00	○ A	◉ B	
		Reset		Continue

Fig. 3 Choice Menu for Eliciting the Sooner Equivalent of a Temporal Prospect. In each row participants had to indicate whether they preferred the respective sooner payment ("Option A") or the later payment of the temporal prospect ("Option B"). The observed sooner equivalent $se(x)$ is calculated as the average of the two sooner payments where a preference switch occurred. In this example, the observed sooner equivalent is EUR 31.35

$$RRP(R) = \frac{ev(R) - ce(R)}{|ev(R)|}\,, \tag{7}$$

which is dimensionless and, thus, allows for direct comparisons between prospects with different outcome magnitudes.

In the following, we focus on observed relative risk aversion.[10] Figure 4 shows the mean relative risk premia of the base and scaled-up prospects. First, the participants' risk attitudes exhibit the typical pattern of probability dependence, which is in line with a regressive probability weighting function, i.e., a function that cuts the diagonal from above. That is, on average participants are risk seeking when the probability of receiving the higher outcome is 10%, they are nearly risk neutral when that probability is 50%, and they are risk averse when it is 90%.

Next, we examine a model-free measure of the magnitude effect in risk taking. Consider two risky prospects, R and R_λ. They share the same probabilities, but the

[10] The corresponding analysis of absolute risk aversion is delegated to Appendix A. This latter data is primarily used to discriminate among different specifications of the utility function, but has no particular significance for our discussion of magnitude effects. As the raw data in the appendix shows, there is some evidence that absolute risk aversion decreases with stake size.

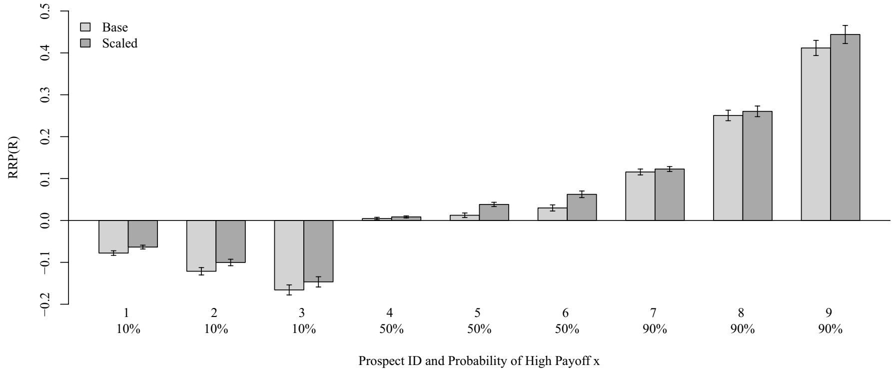


Fig. 4 Mean Relative Risk Premia (RRP) for Base and Scaled-Up Prospects. Mean risk premia for base prospects R and scaled-up prospects R_λ, $\lambda = 5$, for each set of prospects. Error bars indicate the region of +/- one standard error around the mean

outcomes of R are scaled up by a common factor $\lambda = 5$ in R_λ. We define the magnitude effect in risk taking, $M_{rel}(R, \lambda)$, as the change in the relative risk premia due to the scaling up of the outcomes:

$$
\begin{aligned}
M_{rel}(R, \lambda) &:= RRP(R_\lambda) - RRP(R) \\
&= \frac{ev(R_\lambda) - ce(R_\lambda)}{|ev(R_\lambda)|} - \frac{ev(R) - ce(R)}{|ev(R)|} \\
&= \frac{\lambda\, ev(R) - ce(R_\lambda)}{\lambda\, |ev(R)|} - \frac{\lambda\, (ev(R) - ce(R))}{\lambda\, |ev(R)|} \\
&= \frac{\lambda\, ce(R) - ce(R_\lambda)}{\lambda\, |ev(R)|}\,.
\end{aligned}
\tag{8}
$$

When $M_{rel}(R, \lambda) > 0$, relative risk aversion increases with stake size; and when $M_{rel}(R, \lambda) < 0$, relative risk aversion decreases with stake size (Lefebvre et al., 2010).

Table 3 summarizes the magnitude effect for the different risky prospects in our data. It also shows the p-values for one-sided paired Wilcoxon signed-rank tests (*WILC*) and t-tests for the hypothesis of increasing relative risk aversion, i.e., $M_{rel}(R, \lambda) > 0$.

We find strong support for the hypothesis that relative risk aversion increases with stake size. The overall magnitude effect is 0.019 and highly significant. Further tests on subsets of the data consistently confirm increasing relative risk aversion for all probabilities and standard deviations. Figure 5 illustrates these results by showing the relative magnitude effects for prospects with the same probabilities. To summarize, we see a pronounced probability dependence of risk premia, as well as evidence of increasing relative risk aversion.

Table 3 Magnitude Effects in Risk Taking

	Prospects	M_{rel}	*WILC*	$t - Test$
	All	0.019	<0.001	<0.001
p	10%	0.019	0.001	<0.001
	50%	0.022	<0.001	<0.001
	90%	0.015	0.028	0.039
SD	low	0.009	0.001	0.005
	medium	0.020	<0.001	<0.001
	high	0.028	<0.001	<0.001

The average effects of scaling the base prospects' outcomes by $\lambda = 5$ on relative risk aversion (M_{rel}) for all prospects, as well as for prospects sharing the same success probability *p* or standard deviation *SD*. The *p*-values of one-sided paired Wilcoxon Signed-Rank tests and t-tests are reported in the columns "*WILC*" and "$t - Test$", respectively

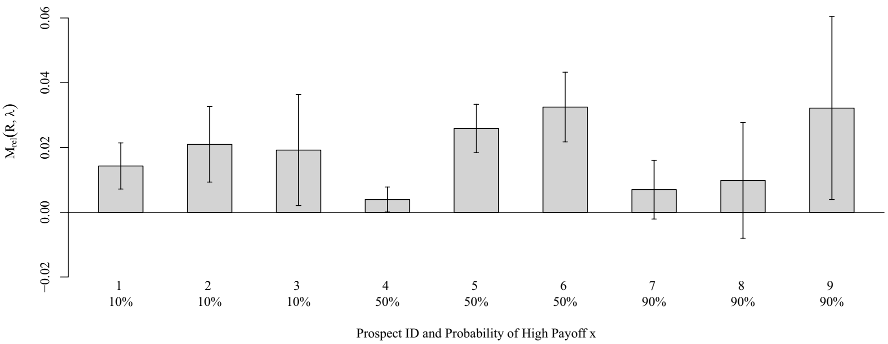


Fig. 5 Mean Magnitude Effects in Risk Taking. Mean magnitude effects when the outcomes of the base prospects are scaled up by a factor of $\lambda = 5$. Error bars indicate the region of +/- one standard error around the mean

4.2 Time discounting data

We now turn to the time discounting data. Analogous to the risk taking data, we employ a model-free characterization of time discounting behavior. In Fig. 6 we report the relative sooner equivalents $RSE(x) := \frac{se(x)}{x}$ for each time delay (t_1, t_2) which provide a measure for the cash discount weights. The bars display the relative sooner equivalents separately for each of the four different levels of future outcomes. Overall, future payments are discounted quite heavily since the heights of the bars are substantially lower than unity. For example, the very first bar on the left indicates that the mean present value of $x = 14$ due in two months is only 64.7% of x, i.e., amounting to 9.054. In other words, a two-month delay leads to a loss in value of 35.3%. This number corresponds to an annual cash discount rate of 261.5% (assuming con-

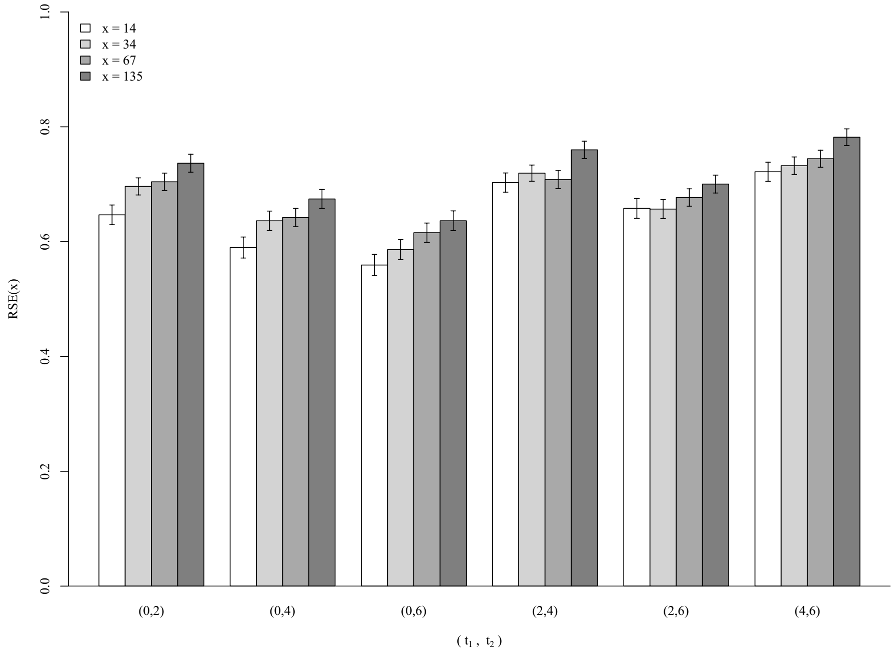


Fig. 6 Mean Relative Sooner Equivalents. The figure shows the mean relative sooner equivalents $RSE(x) = se(x)/x$ for each level of outcome magnitude x and each combination of (t_1, t_2). Error bars indicate the region of +/- one standard error around the mean

tinuous compounding).[11] High discount rates, observed both in experiments and in real purchase decisions, are the rule rather than the exception (Frederick et al., 2002).

Aside from the high level of impatience across all future outcomes and delays, two patterns emerge. First, there is evidence for hyperbolic discounting. Comparing the bars for a given x across the two-month delays, (0, 2), (2, 4) and (4, 6), their heights show a tendency of increasing with t_2, i.e., the loss in value of outcome x over a two-month delay decreases with delay t_2. Second, there is also evidence for the magnitude effect in time discounting. For a given delay (t_1, t_2), we see a tendency of the bar heights to increase with x, i.e., the relative loss of value decreases with outcome magnitude x.

To summarize our descriptive analyses: We find evidence for increasing relative risk aversion as well as hyperbolic discounting and the magnitude effect in discounting of cash payments. Ultimately, we are interested in utility discount rates and not cash discount rates, which differ systematically from each other unless utility is linear. Since utility is not directly observable we need to rely on estimates to examine patterns in utility discount rates. Table 26 in Appendix D shows that the estimated utility discount rates exhibit the same qualitative patterns as the cash rates: High impatience, hyperbolic discounting and magnitude dependence.

[11] Additional examples can be found in Table 26 in Appendix D.

5 Structural estimation

This section describes how we structurally estimate the parameters characterizing aggregate risk taking and time discounting behaviors. The estimation proceeds in two steps. In the first step, it exploits the data on risk taking to infer the shapes of the utility and probability weighting functions. In the second step, it uses the data on time discounting to estimate the parameters governing participants' discount weights.

5.1 Estimating risk taking

As the observed certainty equivalents reveal a clear pattern of probability dependence, a model is called for that captures this phenomenon. Therefore, we opt for Rank Dependent Expected Utility Theory (RDU) which features a probability weighting function. In the first step, therefore, structural estimation applies RDU to account for the patterns in the risk taking data. According to RDU, the decision maker evaluates the risky prospect $R = (x, p; y, 1-p)$ with $x > y \geq 0$ as

$$v(R) = u(x)\, w(p) + u(y)\, [1 - w(p)]\,.$$

We implement the following functional forms for the model components. We use the neo-additive specification for the probability weighting function (Chateauneuf et al., 2007) defined by

$$w(p) = \begin{cases} 0 & \text{for } p = 0 \\ \alpha p + \beta & \text{for } 0 < p < 1 \\ 1 & \text{for } p = 1 \end{cases}\;, \tag{9}$$

for $0 < \beta < 1$ and $0 < \alpha \leq 1 - \beta$. This functional form is a linear approximation of the widely used inverse-S shaped variants and has turned out to fit better here than the standard two-parameter specification based on Prelec (1998).[12] The slope of the curve α measures sensitivity towards changes in probability, the intercept β governs elevation, ceteris paribus. The higher is β, the comparatively higher is risk tolerance.

Our favored specification of the utility function is a one-parameter version of the expo-power hybrid function,

$$u(x) = 1 - \exp(-x^{\gamma}/\gamma). \tag{10}$$

The parameter $\gamma > 0$ describes the utility function's curvature. This specification of the utility function not only fits the data well but also allows for increasing relative risk aversion and decreasing absolute risk aversion at the same time if $\gamma < 1$. To improve the fit of the model and facilitate its estimation, we normalized all outcomes x to lie within the interval [0, 1], i.e., we scaled the outcomes such that 0 is the lowest and 1 is the highest outcome across all risky prospects of the experiment (Abdellaoui et al., 2007; Hastie et al., 2009). We also examined the fit of various widely

[12] Results are available upon request.

used alternative functional forms for the utility function: the power function, the exponential function, and the generalized logarithmic function. Only the latter functional form can accommodate both increasing relative risk aversion and decreasing absolute risk aversion as well. None of these alternative specifications resulted in a superior model fit when compared with the expo-power function. Details are available in Appendix B.

To estimate the parameters of the model, we assume that the decision maker commits a random error, such that the certainty equivalent can be expressed as

$$ce(R) = u^{-1}\left[v(R)\right] + e_R\,,$$

where $u^{-1}\left[v(R)\right]$ is the predicted certainty equivalent, and e_R is the normally distributed error with standard deviation σ_R. Maximizing the corresponding log likelihood function yields the parameter estimates for the shape of the probability weighting function, $\hat{\alpha}$ and $\hat{\beta}$; the curvature of the utility function, $\hat{\gamma}$; and the standard deviation $\hat{\sigma}_R$ of the random error.

5.2 Estimating two-speeds time discounting

In the second step, we estimate the parameters governing the discount weights of the two-speeds model. The data derived from smaller sooner versus larger later outcomes does not allow to separate the curvatures of the utility function and the discount weights. Thus, we use the estimated curvature of the utility function, $\hat{\gamma}$, previously obtained in the first step from the data on risk taking, to identify the discount weights. This procedure was used frequently before (Andersen et al., 2008; Epper et al., 2011) and is consistent with our theoretical approach to explaining the magnitude dependence of the discount weights.

As discussed before, the observed present value of the temporal prospect $T = [x, t]$ is given by

$$V_0(T) = u(x)\,D(t;x)\,, \tag{11}$$

where $D(t;x) = \omega(x)\exp(-\eta t) + (1-\omega(x))\exp(-\theta t)$, $\theta > \eta$, represents the mixture of exponential discount functions. We estimated two competing specifications of $D(t; x)$, the baseline case **2SPEEDS** and a restricted version, **2SPEEDS.R**, which we will comment on later. Additionally, we investigated the extended version, **2SPEEDS.E**, that can accommodate increasing impatience, featuring an additional parameter κ (Table 4).

To estimate the parameters of each of these specifications, we assume that the decision maker makes a normally distributed error, e_T, when she states the sooner equivalent $se(x)$ at t_1 for the later payment (x, t_2). Maximizing the corresponding log likelihood function provides the estimates of the parameters of $D(t; x)$ and the standard deviation of the error, $\hat{\sigma}_T$. Finally, to account for the two-step-nature of the estimation and potential serial correlation, we calculate cluster robust standard errors using the bootstrap at the individual level with 5,000 replications.

Table 4 Two-Speed Models of Discounting Behavior $D(t; x)$

Model	$D(x, t)$	Parameters	Restriction
2SPEEDS	$\frac{u(\delta x)}{u(x)} \exp(-\eta t) + \left(1 - \frac{u(\delta x)}{u(x)}\right) \exp(-\theta t)$	δ, η, θ	
2SPEEDS.R	$\frac{u(\delta x)}{u(x)} + \left(1 - \frac{u(\delta x)}{u(x)}\right) \exp(-\theta t)$	δ, θ	$\eta = 0$
2SPEEDS.E	$\frac{u(\delta x)}{u(x)} \exp(-(\eta t)^{\kappa}) + \left(1 - \frac{u(\delta x)}{u(x)}\right) \exp(-(\theta t)^{\kappa})$	$\delta, \eta, \theta, \kappa$	

6 Results: aggregate behavior

6.1 Structural estimates for risk taking

We first present the structural estimates for risk taking in Table 5. The estimated curvature of the utility function amounts to $\hat{\gamma} = 0.852$. Its graph is depicted on the left of Fig. 7. As expected, the parameters of the probability weighting function, $\hat{\alpha} = 0.364$ and $\hat{\beta} = 0.283$, generate a regressive curve, shown on the right of Fig. 7, which departs comparatively strongly from the diagonal and implies a substantial deviation from the objectively given probabilities. Such a pronounced departure from linear probability weighting is quite unusual. Estimates for Prelec's α, that measures departure from linearity and for which we have comparative values, tend to be in the vicinity of 0.5 or higher in student samples (Fehr-Duda & Epper, 2012; l'Haridon & Vieider, 2019; Stott, 2006), whereas it amounts to 0.270 for our data. l'Haridon and Vieider (2019) find $(\hat{\alpha}, \hat{\beta})$-pairs for Prelec probability weights in 30 countries to be distributed across the board (see their Fig. 3, Greece was not included), with $\hat{\alpha}$ below 0.3 in Nigeria standing out. We can only speculate why we observe such a strong departure from linearity in our Greek student sample. At the time of the experiment, Greece had been undergoing a period of great economic uncertainty caused by its sovereign debt crisis which may have had an effect on people's risk preferences.

6.2 Structural estimates for the two-speeds model

Now we take the estimates for risk taking as given and estimate the parameters of the alternative models of the discount weights $D(t; x)$. Table 6 summarizes the parameter estimates for the **2SPEEDS** and **2SPEEDS.R** models. The first important insight is that $\hat{\theta}$ is actually greater than $\hat{\eta}$, consistent with our modeling assumptions. In the unrestricted **2SPEEDS** model the estimate of $\hat{\eta}$ is not significantly different from zero. Therefore, the model **2SPEEDS.R**, in which η

Table 5 Structural Estimates: Risk Taking

Parameters	Estimates	SE	p-value
$\hat{\gamma}$	0.852	0.026	<0.001
$\hat{\alpha}$	0.364	0.020	<0.001
$\hat{\beta}$	0.283	0.011	<0.001
$\hat{\sigma}_R$	0.181	0.005	<0.001

RDU with expo-power utility and neo-additive probability weights

Log Likelihood: 15608.85

AIC: -31209.69

BIC: -31183.12

Number of observations: 5668

Number of individuals: 159

Cluster robust standard errors SE bootstrapped

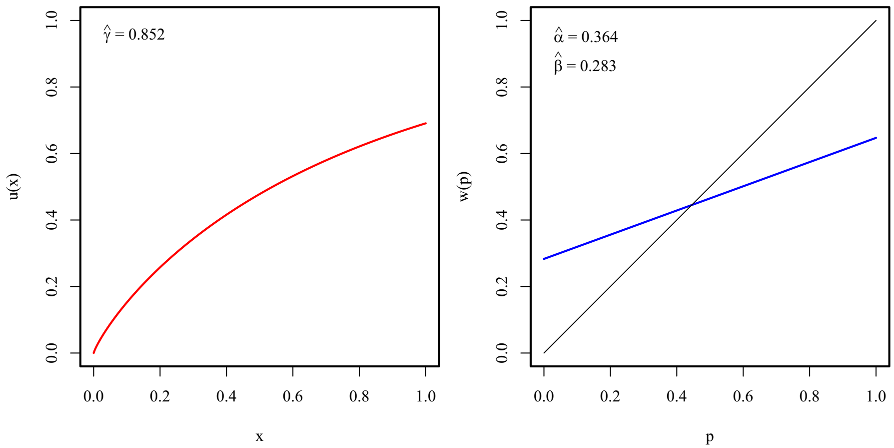


Fig. 7 Estimated Utility and Probability Weights. The left graph shows the estimated utility function $u(x) = -\exp(-x^{\hat{\gamma}}/\hat{\gamma}) + 1$, with $\hat{\gamma} = 0.852$. The right graph depicts the estimated neo-additive probability function with $\hat{\alpha} = 0.364$ and $\hat{\beta} = 0.283$

Table 6 Structural Estimates: Two-Speeds Models

	2SPEEDS			2SPEEDS.R		
Parameters	Estimates	SE	*p*-value	Estimates	SE	*p*-value
$\hat{\delta}$	0.317	0.049	<0.001	0.400	0.018	<0.001
$\hat{\eta}$	-0.219	0.123	0.074			
$\hat{\theta}$	3.276	0.379	<0.001	3.730	0.211	<0.001
$\hat{\sigma}_T$	0.196	0.013	<0.001	0.196	0.013	<0.001
Log Likelihood	4911.107			4910.699		
AIC	-9814.213			-9815.398		
BIC	-9789.248			-9796.675		
Observations	3794			3794		
Individuals	159			159		

Cluster robust standard errors SE bootstrapped

is restricted to zero, is the preferred specification which also exhibits a superior fit according to the *AIC* and *BIC*.

First, we report the estimates for the mixture weights $\hat{\omega}(x)$ in Table 7 which increase with *x* as predicted. As Prelec (2004) noted, the distribution of the mixture weights affects the degree of decreasing impatience of the discount weights - a 60-40 mixture is less decreasingly impatient than a 50-50 one. Thus, outcome magnitude has a direct effect on the degree of hyperbolic decline, which constitutes a special feature of the two-speeds model.

As the raw data revealed high levels of impatience, the estimates of the parameters driving $D(t; x)$ should reflect high impatience, too. With η being zero, there are

Table 7 Mixture Weight Estimates $\hat{\omega}(x)$

x	14	34	67	135
$\hat{\omega}(x)$	0.479	0.503	0.538	0.602

Estimates based on utility curvature $\hat{\gamma} = 0.852$, and depreciation parameter $\hat{\delta} = 0.4$.

two channels through which high impatience may express itself: either through a low depreciation parameter δ and/or through a high discount rate θ. Indeed, the estimated parameters amount to $\hat{\delta} = 0.4$ and $\hat{\theta} = 3.73$. There are several possibilities of rationalizing $\eta = 0$. First, the estimates suggest that participants apply a heuristic to evaluate a future payment x: initially calculating a lower bound $u(\delta x)$, irrespective of the length of delay where δ reflects depreciation for being delayed *per se*, and then adjusting this lower bound by discounting the remaining utility $u(x) - u(\delta x)$ at the rate θt.

The second interpretation links our results to the perception of future uncertainty. With $\eta = 0$, the present value of the future prospect $[x, t]$ can also be written as

$$V_0([x,t]) = u(x)\exp(-\theta t) + u(\delta x)(1 - \exp(-\theta t)),$$

which corresponds to the expected utility of the risky prospect $(x, \exp(-\theta t); \delta x, 1 - \exp(-\theta t))$, where $\exp(-\theta t)$ denotes the subjective delay-dependent probability of receiving x. The longer the delay, the less likely the full reward x is expected to materialize, and the more likely the decision maker will end up with only δx. $\exp(-\theta t)$, therefore, can be viewed as encompassing both, discounting for risk and discounting for delay.

Our model is based on the assumption that the parameter δ is constant. It is of course conceivable that δ depends on some characteristics of the future prospect, for example on outcome magnitude or length of delay. While the lengths of delay in our experiment were not sufficiently variable to test such a hypothesis, we conducted a robustness check concerning outcome magnitudes. We split the temporal prospects into two sets, containing only the smaller payoffs $x \in \{14, 34, 67\}$ and only the larger payoffs $x \in \{34, 67, 135\}$, respectively, and re-estimated the restricted two-speeds time discounting model. Table 8 summarizes the results of this endeavor. The estimates show that the magnitude of $\hat{\delta}$ remains fairly constant with confidence intervals overlapping greatly. Details are available in Appendix C.

Table 8 Robustness of $\hat{\delta}$

Payoffs x	$\hat{\delta}$	SE
All	0.400	0.018
Smaller	0.396	0.018
Larger	0.372	0.020

2SPEEDS.R estimates of δ
based on all payoff levels,
smaller payoffs only,
and larger payoffs only

As the basic version of the two-speeds model already maps decreasing impatience, an additional parameter measuring departure from exponential discounting is not expected to provide much value added over the mixture of exponentials. Nonetheless, for the sake of completeness, we present the estimates for the extended version **2SPEEDS.E** in Appendix C.

6.3 Model estimates for Baucells and Heukamp (2012)

As previously mentioned, Baucells and Heukamp (2012)'s model can accommodate hyperbolic discounting and is, therefore, a suitable competitor for the two-speeds model. Their model was recently put to the test in an experimental investigation by Somasundaram and Eli (2022). The discount function for a linear psychological distance function is given by

$$D(t; x) = \exp(-\rho(x)t),$$

where the discount rate ρ is assumed to depend directly on outcome magnitude. As in Noor (2011), other than $\rho'(x) < 0$ the functional dependence of $\rho(x)$ on x is not specified. Somasundaram and Eli (2022) work with

$$\rho(x) = \rho_0 + M/x,$$

for some constant M and a base discount rate ρ_0. They find that it performs best among several competing specifications. Model estimates for our data are shown in Appendix C. The estimated base discount rate ρ_0 amounts to 107.5%, capturing the high level of impatience in our data. The magnitude parameter M, however, is not statistically distinguishable from zero. Thus, given the pronounced pattern of hyperbolic discounting in our data, this specification is unable to map its structure of magnitude dependence, as is to be expected.

Baucells and Heukamp (2012) also discuss a version that allows for non-exponential discounting of the constant-sensitivity type: $D(t; x) = \exp(-(\rho(x)t)^{\kappa})$, which is more appropriate for our data. We estimated this model with the same functional specification of $\rho(x)$ as above. Model fit improved noticeably but remained considerably inferior to the two-speeds models'. Parameter estimates for ρ_0 amounted to 0.9, and for M again close to zero with substantial departure from exponentiality. However, the estimation procedure encountered numerical problems and failed to generate standard errors.

We also investigated alternative models of magnitude dependence, based on Noor (2011) and Benhabib et al. (2010), using functional specifications previously suggested in the literature. All of these attempts resulted in model fits that are considerably worse than our variants'. Details are presented in Appendix C, with a summary of model fits in Table 21.

7 Results: individual heterogeneity

So far, we have analyzed aggregate behavior. As every experimenter is keenly aware of, participants' behaviors are usually quite heterogeneous and one-size-fits-all models may be mis-specified. Therefore, we investigate risk taking and time discounting at the individual level.

While aggregate time discounting data are characterized by decreasing impatience, there is often a considerable percentage of participants who exhibit the opposite pattern, increasing impatience (e.g. Takeuchi (2011)). Clearly, for these participants a model that imposes decreasing impatience on the data, such as the basic **2SPEEDS** model, would be mis-specified, and **2SPEEDS.E** would be the appropriate model. In our data set, however, the percentage of participants with increasing impatience is negligible. Depending on how strictly one defines consistent behavior in a model-free manner, there are only about 0.6% of the participants with consistently increasing impatience. What we do find, however, is a comparatively large group who cannot be clearly assigned to a consistent discounting type. This group comprises about 20% of the participants. In the following, we focus on the 127 participants with consistent behavior for whom the mixture of exponentials provides an adequate representation. First, we report the aggregate estimation results for the consistent group, which we will complement with estimates at the individual level.

7.1 Structural estimates: consistent participants

Aggregate estimates for the parameters of the utility function and the neo-additive probability weighting function are reported in Table 9 and show no substantial differences to the estimates for the full sample. Figure 8 shows the distributions of the estimated individual parameters. Their median values amount to $\hat{\gamma}_i^{med} = 0.861$, $\hat{\alpha}_i^{med} = 0.365$, and $\hat{\beta}_i^{med} = 0.277$, all of which lie very close to the aggregate estimates in Table 9.

Table 9 Structural Estimates: Risk Taking: Consistent Participants

Parameters	Estimates	SE	*p*-value
$\hat{\gamma}$	0.862	0.029	<0.001
$\hat{\alpha}$	0.346	0.023	<0.001
$\hat{\beta}$	0.289	0.012	<0.001
$\hat{\sigma}_R$	0.185	0.005	<0.001

RDU with expo-power utility and neo-additive probability weights

Log Likelihood: 12337.43

AIC: -24666.86

BIC: -24641.20

Number of observations: 4521

Number of individuals: 127

Cluster robust standard errors SE bootstrapped

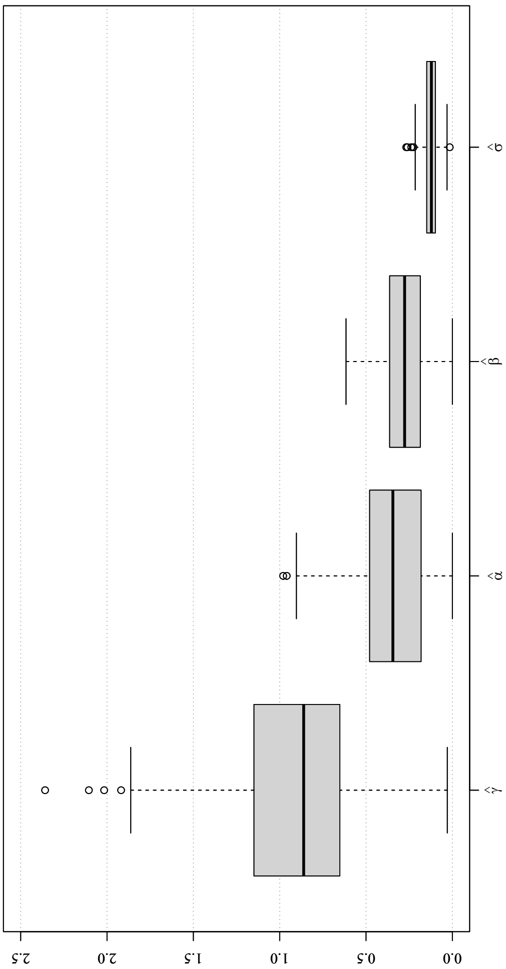


Fig. 8 Distributions of Risk Parameters. The boxplots show the distributions of the individual parameters for the risk model. Estimates for the 127 consistent participants only

Turning to time discounting, in the **2SPEEDS** model, the estimate of η turns out to be indistinguishable from zero, as previously, with a p-value of 0.872. As before, restricting η to zero provides a better fit. Estimates of the corresponding **2SPEEDS.R** model are summarized in Table 10. Not surprisingly, focusing on the consistent participants affects the parameter estimates for δ and θ compared with the full sample estimates. The depreciation parameter δ decreases somewhat, whereas the discount rate θ increases, i.e. overall, estimates for the consistent participants exhibit a more pronounced discounting pattern.

Figure 9 shows the distributions of the individual **2SPEEDS.R** parameter estimates. The median estimates are given by $\hat{\delta}_i^{med} = 0.33$ and $\hat{\theta}_i^{med} = 3.24$ which align fairly well with their aggregate counterparts.

Furthermore, we analyze the relationship between *observed* measures and *estimated* quantities at the individual level. First, a prediction of our model concerns the relationship between the mixture weights and the magnitudes of the discount weights. Recall from the proof of the Proposition that, ceteris paribus, $D(t; x)$ increases with $\omega(x)$. Thus, we should observe a positive correlation between the *estimated* utility mixture weights $\hat{\omega}_i(x)$ and the *observed* cash discount weights $RSE_i(x) = se_i(x)/x$. This conjecture is clearly confirmed by Fig. 10. The regression line is characterized by a highly significant slope amounting to 0.358 (p-value < 0.001).

Finally, if our model has any explanatory power, we should find a positive relationship between individual magnitude effects in discounting and the decrease in the elasticity of the estimated utility functions. We define the *observed individual magnitude effect* in discounting, ME_i, as the difference between individual's i mean relative sooner equivalents for the highest and lowest outcome, $x_{max} = 135$ and $x_{min} = 14$:

$$ME_i := \frac{\overline{se}_i(x_{max})}{x_{max}} - \frac{\overline{se}_i(x_{min})}{x_{min}}. \tag{12}$$

As has become clear from the proof of Proposition 1, an appropriate measure of the decrease in the elasticity is the estimated change in the respective mixture weights $\hat{\omega}_i(x)$. Consequently, we define the change in the mixture weights estimated for individual i in the following way:

$$\Delta\hat{\omega}_i := \frac{u_i(\hat{\delta}_i x_{max})}{u_i(x_{max})} - \frac{u_i(\hat{\delta}_i x_{min})}{u_i(x_{min})}. \tag{13}$$

Plotting $\Delta\hat{\omega}_i$ against ME_i in Fig. 11 shows a positive relationship. The corresponding regression line is characterized by an intercept of -0.41, which is not significantly different from zero (p-value = 0.151), and a slope of 1.37, which is highly significant (p-value < 0.001). Thus, there is a significant positive relationship between *observed*

Table 10 Structural Estimates of **2SPEEDS.R**: Consistent Participants

Parameters	Estimates	SE	p-value
$\hat{\delta}$	0.397	0.020	<0.001
$\hat{\theta}$	4.019	0.245	<0.001
$\hat{\sigma}_T$	0.197	0.015	<0.001

Log Likelihood: 3910.136

AIC: -7814.272

BIC: -7796.221

Number of observations: 3032

Number of individuals: 127

Cluster robust standard errors SE bootstrapped

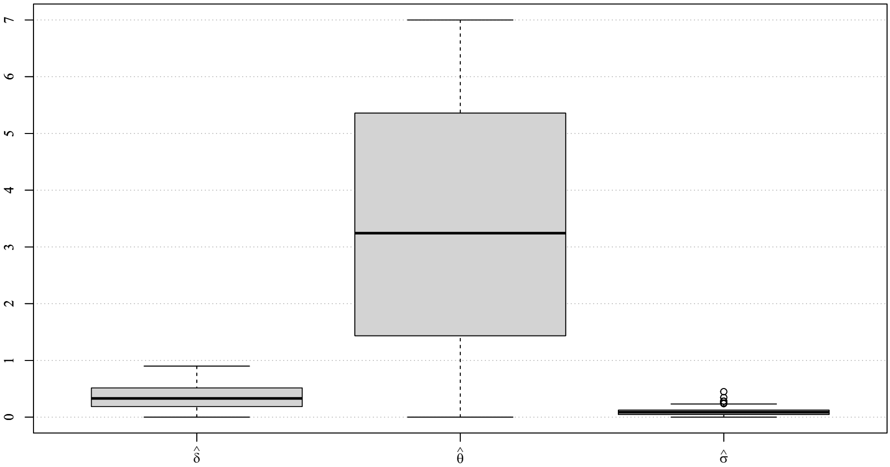


Fig. 9 Distributions of **2SPEEDS.R** Parameters. The boxplots show the distributions of the individual parameters for the discounting model. Estimates for the 127 consistent participants only

magnitude effects in discounting and *estimated* measures of decreasing elasticity at the individual level, which corroborates the within-data predictive power of our model.

7.2 Parameter estimates and individual characteristics

We also investigated whether estimated parameters are related to individual characteristics that we measured in the experimental questionnaire at the end of the sessions. We ran a number of regressions of the estimated parameters on the following individual characteristics:

- MALE = 1 if participant responded accordingly, 0 otherwise

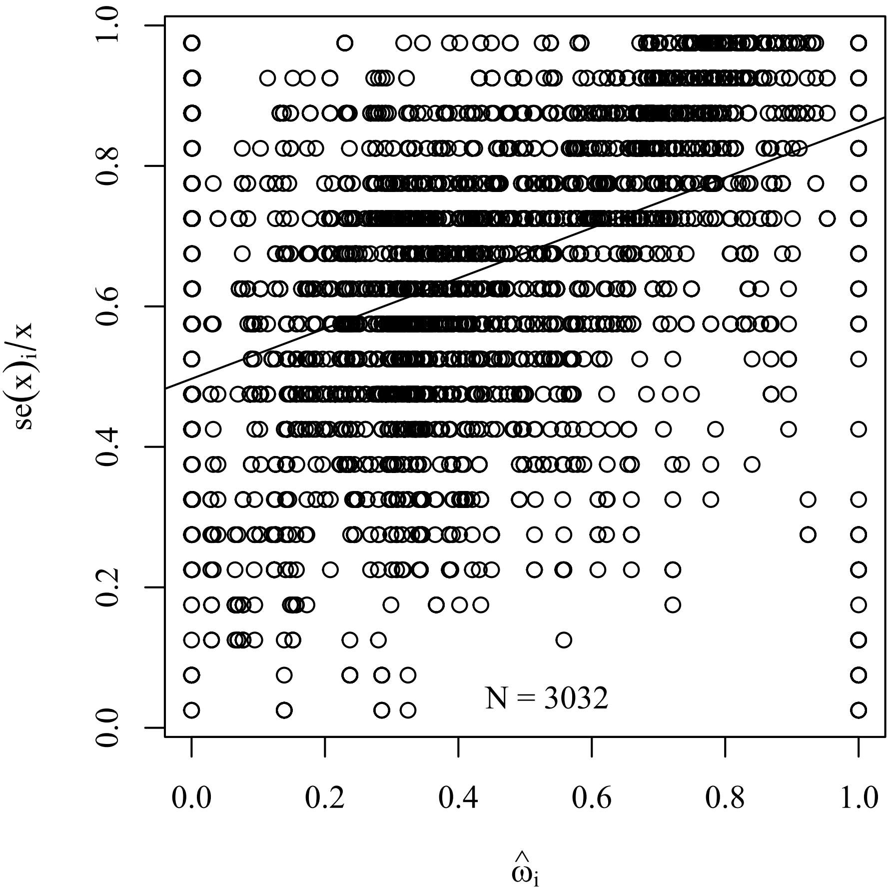


Fig. 10 Individual Mixture Weights and Observed Discount Weights. The circles show the observed individual cash discount weights RSE_i plotted against the respective estimated individual mixture weights, $\hat{\omega_i}$, as well as the linear regression line, characterized by an estimated intercept of 0.497 and a slope of 0.358. Both values are highly significant (p-values < 0.001). Adjusted R-squared amounts to 0.2038. Estimates for the 127 consistent participants only

- ECON = 1 if participants had attended economics lectures before, 0 otherwise
- RISKWILL = 1, if self-reported willingness to take risks – measured on a Likert scale from 1 (low) to 10 (high) – exceeds 5, and 0 otherwise
- PATIENCE = 1, if self-reported patience – measured on a Likert scale from 1 (low patience) to 10 (high patience) – exceeds 5, and 0 otherwise

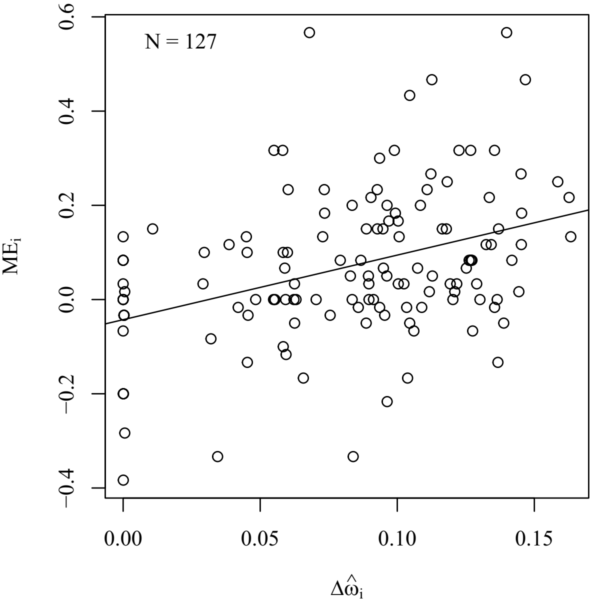


Fig. 11 Individual Magnitude Effects and Decreasing Elasticity. The circles show the observed individual magnitude effects ME_i plotted against the respective changes in estimated individual mixture weights, $\Delta\hat{\omega}_i$, which provide individual measures of decreasing elasticity, as well as the linear regression line, characterized by an intercept of -0.042 (p-value = 0.151) and an estimated slope of 1.37 (p-value < 0.001). Estimates for the 127 consistent participants only

Parameter estimates and their bootstrapped standard errors are presented in Tables 22 to 25 in Appendix C.[13]

Turning to the risk model parameters first, we do not find a relationship between utility curvature $\hat{\gamma}$ and any of the variables considered. Furthermore, no gender differences were detected in the probability weighting parameters $\hat{\alpha}$ and $\hat{\beta}$. Although there are many studies concluding that women tend to be more risk averse than men, Niederle (2016)'s literature review finds quite a few where no difference was evident. ECON has a positive influence on the slope of the probability weighting curve α, rendering these participants somewhat more rational (parameter estimate 0.136 with SE = 0.049 bootstrapped), as expected.

For the domain of time discounting, there is no significant relationship between $\hat{\delta}$ and any of the considered variables. Concerning $\hat{\theta}$, there is a significant gender difference in the discount rate with men being substantially more impatient (parameter estimate 0.988 with SE = 0.433 bootstrapped), consistent with the experimental results by Dittrich and Leipold (2014), for example, although there is also evidence to the contrary (Falk et al., 2018).

[13] Bootstrapped standard errors take into account that the dependent variable is an estimated entity.

8 Discussion

Previous attempts at harmonizing utility under risk and over time have faced the difficulty that the observed patterns of behavior seem to call for opposing characteristics of the utility function (Prelec & Loewenstein, 1991). Therefore, extant solutions of the magnitude paradox locate the magnitude effect of discounting not in the utility function but in the discount weights instead. In Baucells and Heukamp (2012)'s model magnitude dependence of discounting is generated by the assumption of discount weights increasing in outcome magnitude where the discount weights are not systematically related to the characteristics of the utility function. We complement this literature by positing an organic link between the characteristics of the utility function and the characteristics of the discount weights.

In our magnitude-dependent two-speeds model, this tight link is generated by the dependence of the mixture weights of the discount functional on the elasticity of the utility function. The crucial insight of our paper states that a utility function accommodating increasing relative risk aversion directly implies the magnitude effect in the discount weights. In this sense, our modeling approach, accompanied by supporting empirical estimates, suggests the unity of the utility function for money over the risk and time domains.

Our approach sheds new light on discounting behavior. We find that the smaller of the two discount rates, η, in the two-speeds model is effectively zero. We offer two potential rationales for this result. The first one interprets participants' behavior as an application of a heuristic: Initially, participants depreciate the promised reward *x* by δ for being delayed *per se* and then they adjust $u(\delta x)$ by discounting the remaining utility difference $u(x) - u(\delta x)$ by $\exp(-\theta t)$ according to the length of delay. The second interpretation links our results to the perception of future uncertainty. Participants may view the promised reward as risky - the longer the delay, the less likely the full reward *x* is expected to materialize, and the more likely the decision maker will end up with only a fraction of *x*. The subjective delay-dependent probability of *x*, $\exp(-\theta t)$, can thus be viewed as encompassing both, discounting for risk and discounting for delay.

Our results may seem surprising. However, neuroscientific research has uncovered a similar pattern. In their study on time discounting of primary rewards, such as juice and water, McLure et al. (2007) fit a (magnitude-independent) mixture of exponentials $D(t)$ to their data and also find that the smaller of the discount rates is statistically indistinguishable from zero. They attribute their findings to the differential levels of impatience of two distinct areas in the brain. They find that activity in the limbic reward system declines rapidly as rewards are delayed and identify this neural system as the comparatively impatient θ-system (in our notation). In contrast, the frontal/parietal system seems to be much less sensitive to the timing of rewards. Consequently, the authors characterize this neural system as the comparatively patient η-system. For the latter system, there is also a good correspondence between their estimates of $D(t)$ and the neural activation estimates. Of course, it needs to be seen whether such a pattern of neural activation can also be observed in magnitude-dependent discounting.

In our analysis we assumed, for reasons of parsimony, δ to be constant. Of course, depreciation might follow a more complex pattern, possibly depending on outcome magnitude and/or time horizon. Regarding dependence on outcome magnitude robustness checks show that the estimates of δ remain quite stable. With respect to delay dependence our data set does not contain sufficient variability in the lengths of delay. To explore this possibility is a task for future research requiring new richer data sets.

Our results, as any experimental findings, depend on a number of factors: participant pool, choice set, elicitation method, and potentially also environmental conditions. When we conducted the experiment Greece had been experiencing difficult economic times caused by its sovereign debt crisis which might have had an effect on people's preferences. It is, therefore, likely that estimated parameters turn out quite different in experiments with different participant pools and different macroeconomic backgrounds.

With the exception of magnitude-dependent exponential discounting, our model can accommodate a wide range of differing behaviors whenever the utility function is characterized by decreasing elasticity. Our raw data shows not only magnitude dependence both in risk taking and time discounting, it also exhibits pronounced hyperbolic discounting. It does not come as a big surprise that our model fits considerably better than specifications that retain exponential discounting. Hyperbolic discounting need not be the predominant pattern in other data sets, however (e.g. Imai et al. (2021)). If discounting is essentially exponential and there is a significant magnitude effect present in the data, models such as Noor (2011)'s may be the appropriate choice. One difficulty with the models suggested by Noor (2011) and Baucells and Heukamp (2012) lies in their generality - they do not prescribe specific functional forms, which poses a challenge for empirical work. It is therefore conceivable that Baucells and Heukamp (2012)'s non-exponential version of the model performs better under parameterizations other than the previously tested ones. For data sets that feature both non-exponential discounting and the magnitude effect, the two-speeds model is straightforward to implement and provides a useful addition to the toolbox of the experimenter. In any case, when there is heterogeneity in the population it may be advisable to work with an appropriate finite mixture model or similar methods if individual estimates are not feasible or desirable.

Finally, our results rely on the estimated utility function and, thus, a crucial assumption concerns the stability of probability weights across outcome magnitudes. There is evidence stemming from experiments with Chinese participants that probability weights react substantially to high outcomes (Fehr-Duda et al., 2010; Kachelmeier & Shehata, 1992). However, Bouchouicha and Vieider (2017) do not find such an effect in their data. In our data set, we can reject magnitude dependence of the probability weights as well (Hofland, 2022). Even though our Greek data set is exceptionally rich, stakes were considerably lower than the ones employed in the Chinese experiments. Thus, we cannot preclude that probability weights do react to increasing stakes when they are sufficiently high. This issue is another interesting question to be explored in future work.

Appendix A: Analysis of absolute risk aversion

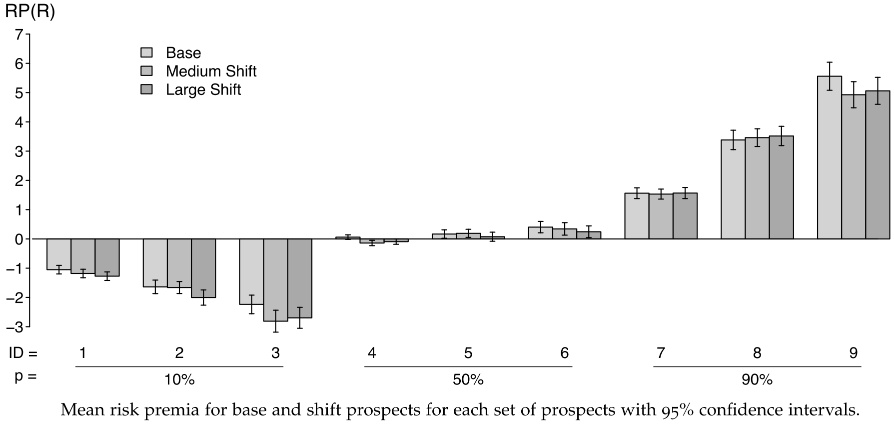


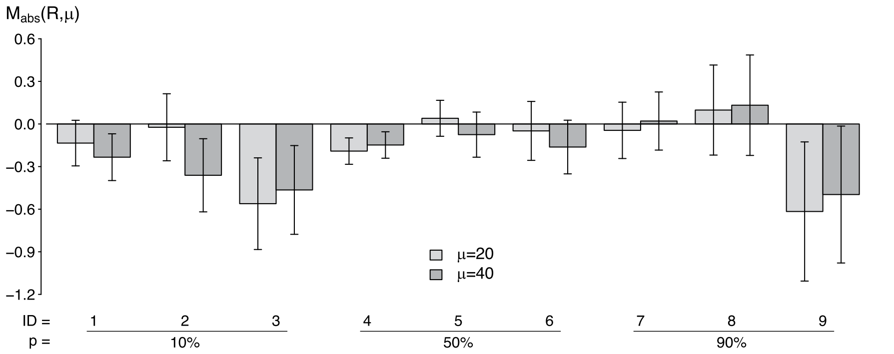
Fig. 12 Mean Risk Premia (RP) for Base and Shift Prospects. Mean risk premia for base and shift prospects for each set of prospects with 95% confidence intervals

We start with measuring the absolute magnitude effect. Consider two risky prospects, R and R_{μ}. They have the same probabilities, but all outcomes of R_{μ} are higher than those of R by an absolute amount $\mu > 0$ (Farquhar & Nakamura, 1987). The change in risk premia due to this shift in outcomes is the absolute magnitude effect on risk aversion:

$$\begin{aligned} M_{abs}(R,\mu) &= RP(R_{\mu}) - RP(R) \\ &= ev(R_{\mu}) - ce(R_{\mu}) - (ev(R) - ce(R)) \\ &= ev(R) + \mu - ce(R_{\mu}) - (ev(R) - ce(R)) \\ &= ce(R) + \mu - ce(R_{\mu}) \,. \end{aligned} \tag{A1}$$

The sign of $M_{abs}(R,\mu)$ reflects the direction of the absolute magnitude effect. That is, when $M_{abs}(R,\mu) > 0$, absolute risk aversion increases with stake size; and when $M_{abs}(R,\mu) < 0$, it decreases with stake size.

Overall, we find support for the hypothesis that absolute risk aversion decreases with stake size. The average absolute magnitude effects on risk aversion are $M_{abs}(R,\mu = 20) = -0.16$ and $M_{abs}(R,\mu = 40) = -0.20$, confirming decreasing absolute risk aversion. Additional tests on subsets of the data reveal decreasing absolute risk aversion for the prospects with $p = 10\%$ and $p = 50\%$ as well as for those with low and high standard deviations. However, there is no support for absolute risk aversion decreasing between the medium and large shift prospects. Thus, decreasing absolute risk aversion appears to become weaker for larger stakes. Figure 13 illustrates the absolute magnitude effects for prospects sharing the same probabilities p. The decline in absolute risk aversion is most consistent for prospects with $p = 10\%$ and a high standard deviation, as is visible in Table 11.



Absolute magnitude effects for $\mu = 20$ and $\mu = 40$ increases in base prospect outcomes for each set of prospects with 95% confidence intervals.

Fig. 13 Absolute Magnitude Effects for 20 and 40 EUR Shifts. Absolute magnitude effects for $\mu = 20$ and $\mu = 40$ increases in base prospect outcomes for each set of prospects with 95% confidence intervals

Table 11 Absolute and Relative Magnitude Effects

		$\mu = 20$			$\mu = 40$			$\lambda = 5$		
	Prospects	M_{abs}	WILC	t-Test	M_{abs}	WILC	t-Test	M_{rel}	WILC	t-Test
All		-0.167	<0.001	0.001	-0.204	<0.001	<0.001	0.019	<0.001	<0.001
p	10%	-0.256	<0.001	<0.001	-0.369	<0.001	<0.001	0.019	<0.001	<0.001
	50%	-0.073	0.065	0.055	-0.139	0.001	0.003	0.022	<0.001	<0.001
	90%	-0.173	0.037	0.073	-0.105	0.198	0.189	0.015	0.028	0.039
SD	low	-0.131	0.004	0.002	-0.124	0.007	0.009	0.009	0.001	0.005
	medium	0.042	0.771	0.720	-0.110	0.140	0.094	0.020	<0.001	<0.001
	high	-0.409	<0.001	<0.001	-0.373	<0.001	<0.001	0.028	<0.001	<0.001

Average effects of $\mu = 20$ and $\mu = 40$ shifts and a factor $\lambda = 5$ increase in base prospect outcomes on absolute risk aversion (M_{abs}) and relative risk aversion (M_{rel}) for all prospects and for prospects sharing the same success probability p or standard deviation *SD*. The p-values of one-sided paired Wilcoxon Signed-Rank tests and t-tests are reported in columns "WILC" and "t-Test"

Appendix B: Alternative specifications of utility

Aside from the hybrid expo-power specification discussed in the main text, we investigated the fit of several other functional forms for the utility function for risk, all of which are characterized by a single parameter γ. In all the cases we assumed the neo-additive specification of probability weighting which had turned out to provide a better fit than the commonly used functional form introduced by Prelec (1998).

Specifications

1. **The Power Utility Function**

$$u(x) = \begin{cases} \frac{x^{1-\gamma}}{\gamma} & \text{if } \gamma < 1 \\ \ln(x) & \text{if } \gamma = 1 \\ -\frac{x^{1-\gamma}}{\gamma} & \text{if } \gamma > 1 \end{cases}$$

2. **The Exponential Utility Function**

$$u(x) = \begin{cases} \frac{1-\exp(-\gamma x)}{\gamma} & \text{if } \gamma \neq 0, \\ x & \text{if } \gamma = 0 \end{cases}$$

3. **The Generalized Logarithmic Function**

$$u(x) = \begin{cases} \frac{\ln(1+\gamma x)}{\gamma} & \text{if } \gamma \neq 0, \\ x & \text{if } \gamma = 0 \end{cases}$$

Estimation results

1. **The Power Utility Function**

Table 12 RDU with Power Utility

Parameters	Estimates	SE	P-value
$\hat{\gamma}$	0.280	0.024	<0.001
$\hat{\alpha}$	0.370	0.006	<0.001
$\hat{\beta}$	0.271	0.004	<0.001
$\hat{\sigma}_R$	0.181	0.002	<0.001

Log Likelihood: 15585.21

AIC: -31162.42

BIC: -31135.85

Number of observations: 5668

Number of individuals: 159

Standard errors clustered on individuals

2. **The Exponential Utility Function**

Table 13 RDU with Exponential Utility

Parameters	Estimates	SE	P-value
$\hat{\gamma}$	1.803	0.165	<0.001
$\hat{\alpha}$	0.352	0.006	<0.001
$\hat{\beta}$	0.288	0.005	<0.001
$\hat{\sigma}_R$	0.181	0.002	<0.001

Log Likelihood: 15555.42

AIC: -31198.79

BIC: -31172.22

Number of observations: 5668

Number of individuals: 159

3. **The Generalized Logarithmic Function**
Among these variants the generalized logarithmic function performs best but is inferior to the hybrid expo-power function shown in Table 5 in the main text.

Table 14 RDU with Logarithmic Utility

Parameters	Estimates	SE	P-value
$\hat{\gamma}$	4.533	0.764	<0.001
$\hat{\alpha}$	0.364	0.006	<0.001
$\hat{\beta}$	0.283	0.005	<0.001
$\hat{\sigma}_R$	0.181	0.002	<0.001

Log Likelihood: 15604.46

AIC: -31200.91

BIC: -31174.34

Number of observations: 5668

Number of individuals: 159

Appendix C: Additional analyses

C.1: Robustness checks on $\hat{\delta}$

Table 15 Structural Estimates of **2SPEEDS.R**, Smaller Payoffs Only

Parameters	Estimates	SE	*p*-value
$\hat{\delta}$	0.396	0.018	<0.001
$\hat{\theta}$	3.766	0.225	<0.001
$\hat{\sigma}_T$	0.218	0.015	<0.001

Based on observations for $x \in \{14, 34, 67\}$

RDU with expo-power utility and neo-additive probability weights

Log Likelihood: 4406.984

AIC: -8807.969

BIC: -8790.111

Number of observations: 2843

Number of individuals: 159

Cluster robust standard errors SE bootstrapped

Table 16 Structural Estimates of **2SPEEDS.R**, Larger Payoffs Only

Parameters	Estimates	SE	*p*-value
$\hat{\delta}$	0.372	0.020	<0.001
$\hat{\theta}$	2.829	0.202	<0.001
$\hat{\sigma}_T$	0.156	0.008	<0.001

Based on observations for $x \in \{34, 67, 135\}$

RDU with expo-power utility and neo-additive probability weights

Log Likelihood: 3220.999

AIC: -6435.997

BIC: -6418.129

Number of observations: 2853

Number of individuals: 159

Cluster robust standard errors SE bootstrapped

Table 17 Structural Estimates: Extended Model **2SPEEDS.E**

Parameters	Estimates	SE	p-value
$\hat{\delta}$	0.375	0.024	<0.001
$\hat{\eta}$	0.003	0.035	0.929
$\hat{\theta}$	3.390	0.257	<0.001
$\hat{\kappa}$	0.092	0.004	<0.001
$\hat{\sigma}_T$	0.196	0.046	<0.001

RDU with expo-power utility and neo-additive probability weights

Log Likelihood: 4911.362

AIC: -9812.724

BIC: -9781.518

Number of observations: 3794

Number of individuals: 159

Cluster robust standard errors SE bootstrapped

C.2: Parameter estimates for 2SPEEDS.E

As in the basic case **2SPEEDS**, Table 17 shows that the parameter estimate of η turns out to be indistinguishable from zero, and κ is estimated to be close to zero.

As the values of *AIC* and *BIC* indicate, **2SPEEDS.E** hardly provides an improvement over the basic version **2SPEEDS**. Moreover, it is not to be preferred over **2SPEEDS.R**, in line with expectations.[14] **2SPEEDS.E** is a useful specification for individual estimates or in a finite mixture model when there is a substantial number of participants exhibiting increasing impatience. In our data, however, the percentage of participants with increasing impatience is negligible.

C.3: Alternative magnitude-dependent discounting models

1. The Model of Noor (2011), **NOOR**
   Noor (2011) provides axiomatic foundations for a magnitude-dependent discounting model by weakening the classical stationarity axiom of Discounted Utility Theory. He derives a representation $V_0([x,t]) = u(x)D(t;x) = u(x)d(x)^t$, where the discount factor $d(x)^t$ depends on outcome magnitude with $d'(x) > 0$. Noor (2011) proposes the following functional form which we use for structural estimation:

$$d(x) = \rho^{x^{-\xi}},$$

   where the constant discount factor ρ and the magnitude-dependence parameter ξ are to be estimated. Assuming expo-power utility with parameter $\hat{\gamma} = 0.852$ yields the estimates displayed in Table 18.

[14] The bootstrap procedure for **2SPEEDS.E** converged in only 3 out of 5,000 replications which indicates that parameter estimates are at the edge of identifiability.

Table 18 NOOR

Parameters	Estimates	SE	p-value
$\hat{\xi}$	0.156	0.021	<0.001
$\hat{\rho}$	0.415	0.025	<0.001
$\hat{\sigma}_T$	0.202	0.013	<0.001

Log Likelihood: 4798.529

AIC: -9591.058

BIC: -9572.335

Number of observations: 3794

Number of individuals: 159

Cluster robust standard errors SE bootstrapped

Attempts to use power utility, as suggested by Noor (2011), instead of expo-power resulted in a very bad fit.[15] Comparing **NOOR** with our main candidates yields the following ranking: 2SPEEDS.R $\succ$ 2SPEEDS $\succ$ NOOR, with **NOOR** fitting much worse than the other two. The class of generalized exponential discount functions, at least with this specific parametrization, seems to be too restrictive to adequately capture discounting behavior in our data which clearly exhibits a hyperbolic pattern. A simple way to uncover magnitude-dependent *exponential* discounting in experimental data is based on the elicitation of *delay equivalents, de(t)*, instead of sooner equivalents as in our data set. Such an experiment may proceed in the following way:

- Fix two outcomes *x* and *y* with $x < y$.
- Elicit delay equivalents *de*(*t*) by varying *t*, such that $(x, t) \sim (y, de(t))$.
- Check if the delay equivalents are a linear function of *t*: $de(t) = a + bt$. If *de*(*t*) is not linear then exponential discounting can be rejected.
- For *de*(*t*) linear in *t*, a classical magnitude effect is present if the slope $b > 1$. If $b = 1$ there is no magnitude dependence.
- Repeat for different pairings of (*x*, *y*).

To see why this check list makes sense, consider the following derivation. For $(x, t) \sim (y, de(t))$, $u(x)\delta(x)^t = u(y)\delta(y)^{de(t)}$ holds, where $\delta(.)$ is the potentially magnitude-dependent discount factor. Solving for *de*(*t*) results in

$$de(t) = \frac{\ln(u(x)) - \ln(u(y))}{\ln(\delta(y))} + \frac{\ln(\delta(x))}{\ln(\delta(y))}t. \tag{A2}$$

Thus, the intercept is given by $a = \frac{\ln(u(x))-\ln(u(y))}{\ln(\delta(y))}$ and the slope is given by $b = \frac{\ln(\delta(x))}{\ln(\delta(y))}$. It follows that there is no magnitude dependence if $\delta(x) = \delta(y)$, and a classical magnitude effect is present when $\delta(x) < \delta(y)$, i.e., $\frac{\ln(\delta(x))}{\ln(\delta(y))} > 1$ as $\delta(.) < 1$.

2. The Models of Baucells and Heukamp (2012), **BH.L** and **BH.C.**

[15] We also tried a number of alternative specifications of *d*(*x*). These endeavors generally resulted in inferior model fits as well. Details are available upon request.

The model by Baucells and Heukamp (2012) was recently put to the test in an experimental investigation by Somasundaram and Eli (2022). The discount function in the exponential version of this model is defined by

$$D(t; x) = \exp(-\rho(x)t),$$

where the psychological distance function is linear and the discount rate ρ is assumed to depend directly on outcome magnitude. As in Noor (2011), other than $\rho'(x) < 0$ the functional dependence of $\rho(x)$ on x is not specified. Somasundaram and Eli (2022) worked with

$$\rho(x) = \rho_0 + M/x,$$

for some constant M and a base discount rate ρ_0. They find that it works best among several competing specifications. As Table 19 indicates, for our data, this specification **BH.L** clearly fits even worse than **NOOR**. The estimated base discount rate amounts to 107.5%, capturing the high level of impatience in our data. The magnitude parameter M, however, is not distinguishable from zero. Thus, it appears that this specification is unable to map the structure of magnitude dependence in our data.

Baucells and Heukamp (2012) also allow the psychological distance function to be concave which results in non-exponential discounting of the constant-sensitivity type: $D(t; x) = \exp(-(\rho(x)t)^{1-\kappa})$, which should be more appropriate for our data. We estimated this model, **BH.C**, with the same functional specification of $\rho(x)$ as above. Model fit improved noticeably ($AIC = -9686.841$, $BIC = -9661.876$), surpassing **NOOR** as well, but remained considerably inferior to the two-speeds models'. Parameter estimates for ρ_0 amounted to 0.9, and for M again close to zero with substantial departure from exponentiality. However, the estimation procedure encountered numerical problems and failed to generate standard errors.

Table 19 **BH.L**

Parameters	Estimates	SE	p-value
$\hat{\rho}_0$	1.075	0.101	<0.001
$\hat{M}$	0.006	0.014	0.670
$\hat{\sigma}_T$	0.204	0.013	<0.001

Log Likelihood: 4758.742

AIC: -9511.484

BIC: -9492.760

Number of observations: 3794

Number of individuals: 159

Cluster robust standard errors SE bootstrapped

Table 20 BBS

Parameters	Estimates	SE	P-value
$\hat{\alpha}$	0.897	0.013	<0.001
$\hat{\beta}$	0.017	0.001	<0.001
$\hat{\rho}$	0.283	0.027	<0.001
$\hat{\sigma}_T$	0.213	0.008	<0.001

Log Likelihood: 2315.892

AIC: -4623.783

BIC: -4601.576

Number of observations: 1904

Number of individuals: 159

Standard errors SE clustered on individuals

Table 21 Ranking of Alternative Models according to *AIC* and *BIC*

Model	*AIC*	*BIC*
2SPEEDS.R	-9815.398	-9796.675
2SPEEDS.E	-9814.795	-9789.831
2SPEEDS	-9814.213	-9789.248
BH.C	-9686.841	-9661.876
NOOR	-9591.058	-9572.335
BH.L	-9511.484	-9492.760

All model estimates based on expo-power

utility with curvature $\hat{\gamma} = 0.852$

3. The Model of Benhabib et al. (2010), **BBS**
   Benhabib et al. (2010) suggest a somewhat ad-hoc specification of the discount function:

$$D(t; x) = -\beta/x + \alpha \exp(-\rho t),$$

   which features two additional parameters α and β besides the time preference rate ρ. Unfortunately, estimation is only feasible for genuine *present equivalents*, unless one of the parameters α or β is fixed. Therefore, estimation results are not comparable to the other specifications tested. Assuming expo-power utility with $\hat{\gamma} = 0.852$ again renders the estimation results presented in Table 20. Table 21 provides an overview of the respective model fits for comparable alternatives.

C.4: Parameter estimates and individual characteristics

We regressed the estimated parameters for the risk taking and the time discounting model **2SPEEDS.R** on various individual characteristics of the consistent par-

Table 22 Regression of $\hat{\alpha}$ on Individual Characteristics

Variables	Estimates	SE	p-value
Intercept	0.235	0.057	< 0.001
MALE	0.007	0.044	0.864
ECON	0.136	0.049	0.005
PATIENCE	-0.035	0.044	0.425
RISKWILL	0.041	0.044	0.345

Standard errors SE bootstrapped, 5000 replications

Table 23 Regression of $\hat{\beta}$ on Individual Characteristics

Variables	Estimates	SE	p-value
Intercept	0.264	0.033	< 0.001
MALE	0.011	0.024	0.641
ECON	-0.027	0.031	0.383
PATIENCE	0.063	0.024	0.009
RISKWILL	-0.003	0.024	0.900

Standard errors SE bootstrapped, 5000 replications

Table 24 Regression of $\hat{\delta}$ on Individual Characteristics

Variables	Estimates	SE	p-value
Intercept	0.423	0.074	<0.001
MALE	0.006	0.047	0.894
ECON	-0.026	0.066	0.692
PATIENCE	-0.050	0.046	0.279
RISKWILL	-0.052	0.046	0.265

Standard errors SE bootstrapped, 5000 replications

Table 25 Regression of $\hat{\theta}$ on Individual Characteristics

Variables	Estimates	SE	p-value
Intercept	3.881	0.698	<0.001
MALE	0.988	0.433	0.022
ECON	-0.655	0.619	0.289
PATIENCE	-0.517	0.443	0.243
RISKWILL	-0.212	0.442	0.632

Standard errors SE bootstrapped, 5000 replications

ticipants the data on which we collected in the questionnaire ensuing the decision tasks.[16] The most interesting ones turned out to be

- MALE = 1 if participant responded accordingly, 0 otherwise
- ECON = 1 if participants had attended economics lectures before, 0 otherwise

[16] Distributions of respective responses are available on request.

- RISKWILL = 1, if self-reported willingness to take risks – measured on a Likert scale from 1 (low) to 10 (high) – exceeds 5, and 0 otherwise. Actual wording: “How would you describe yourself: Are you someone who tries to avoid risk or who is willing to take risk?"
- PATIENCE = 1, if self-reported patience – measured on a Likert scale from 1 (low patience) to 10 (high patience) – exceeds 5, and 0 otherwise. Actual wording: “How well does the following statement describe you: I abstain from things today so that I will be able to afford more tomorrow."

Appendix D: Cash and utility discount rates

When utility is not linear, observed discount rates, based on cash amounts, overstate the underlying utility discount rates. The following Table 26 gives some examples of cash discount rates and utility discount rates, assuming continuous compounding:

$$se = x \exp(-\rho_c t),$$

where ρ_c denotes the cash discount rate, and

$$u(se) = u(x) \exp(-\rho_u t),$$

where ρ_u denotes the utility discount rate. Assuming expo-power utility with parameter $\hat{\gamma} = 0.852$ and using the mean observed sooner equivalents, the estimated utility discount rates $\hat{\rho}_u$ always lie substantially below the corresponding cash discount rates ρ_c, but show the same patterns of hyperbolic discounting and magnitude dependence.

Finally, we display the mean relative sooner equivalents with *se*(*x*) and *u*(*x*) transformed by the estimated individual utility functions. Figure 14 shows the same qualitative patterns as Fig. 6 in the main text.

Table 26 Discount Rates Compared

(t_1, t_2)	(0, 2)	(0, 6)	(0, 2)	(0, 6)
x	14	14	135	135
se(*x*)	9.054	7.830	99.446	85.913
ρ_c	261.5%	116.2%	183.4%	90.4%
$\hat{\rho_u}$	207.3%	92.5%	89.3%	45.6%

The table exhibits cash discount rates ρ_c and utility discount rates $\hat{\rho_u}$ for expo-power utility with parameter $\hat{\gamma} = 0.852$, based on the mean observed sooner equivalents *se*(*x*) for various outcome levels x and delays t_2

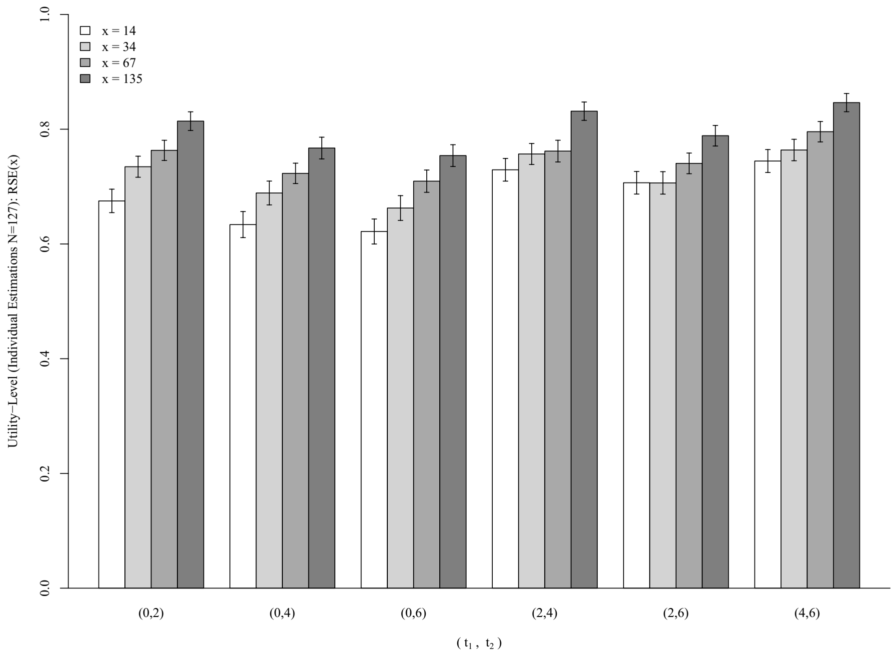


Fig. 14 Relative Sooner Equivalents in Utility Terms. The figure shows the mean relative sooner equivalents in utility terms $u_i(se(x))/u_i(x)$, based on individual utility curvatures $\hat{\gamma}_i$ of the expo-power function, for each level of outcome magnitude x and each combination of (t_1, t_2). Error bars indicate the region of +/- one standard error around the mean

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Author Contributions Authors are listed in the order of their relative contributions with the first two having contributed equally.

Data Availability [The data and a replication package will be made available on a general repository upon acceptance of the paper.]

Declarations

Competing interests The authors declare no competing interests.

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